

Rule-Rewriting Cellular Automata and the Edge of Chaos

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Abstract

We study a two-layer cellular automaton in which each cell carries both a phenotype state and an eight-bit rule. A permutation σ of the eight displayed truth-table positions defines an operator P_σ on those rules by $\text{bit}[\sigma(p)] := \neg \text{bit}[p]$; the resulting family of $8! = 40,320$ permutation-propagators admits exact classification rather than sampling. P_σ has a fixed rule exactly when every cycle of σ is even, an all-even permutation with k cycles has 2^k of them, and every fixed rule is balanced, with Langton activity $\lambda = 1/2$. A coordinate taking one value across every fixed point cannot discriminate among them, and a finite-size scan finds a broad crossover rather than a critical point, so the classical parameter-space reading is closed off inside this family. We substitute a causal measurement—how far a single-cell perturbation travels—and find it governed by how much of what the genotype reads is causally current. Cutting the phenotype-to-genotype edge roughly halves reach; a gain varying only that currency, with spatial statistics matched, moves it smoothly; and sustained rule diversity, mutation, refuges, niches and rule mobility do not. Making the coupling endogenous recovers the same coordinate: a predicate blind to the phenotype reproduces the dial, one reading it exceeds the dial, and one reading a frozen phenotype of matched width and firing rate does not. Posed from inside, currency becomes a limit on regulation—reach is computed from a twin run and so is unavailable to the system, and internal observables predict an intrinsic objective at $R^2 = 0.73$ against 0.02 for reach. No

self-driving search was obtained, and an objective scoring a fixed window ranks configurations inverted against whether they survive; what separates them is persistence. This measure is one of causal propagation and not of dynamical regime.

Keywords: cellular automata, rule evolution, Langton's lambda, self-modifying systems, edge of chaos, artificial life, permutation operators, exhaustive classification, damage spreading, perturbation analysis

Three Senses of the Edge of Chaos

In a cellular automaton whose cells carry rules as well as states, how far a perturbation travels is governed by how much of what the rewriting layer reads is causally current, and not by any parameter of the rules themselves. This paper establishes that coordinate by measurement from outside the system, recovers it from a construction that sets its own coupling, and then poses the same constraint from inside, where it becomes a limit on what the system can regulate: a controller can act only on what it can compute about itself.

The object of study is a finite family of rule-rewriting operators admitting exact classification rather than sampling. A permutation σ of the eight truth-table positions of an elementary rule defines an operator P_σ that rewrites a rule by negating the bits σ names; there are $8! = 40,320$ such operators, and they form the bijective core of the 8^8 possible writings. The family was reached by accident. In 2015, working on a different problem, we built a cellular automaton whose cells carried rules rather than states (Corneli & Maclean, 2015); a programming error connected with one example generated spacetime diagrams reminiscent of edge-of-chaos behaviour in classical CAs, and how that example recovered the behaviour remained open. That example, recreated in Figure 3, is a non-bijective writing outside the core; locating it placed the core itself in view.

“The edge of chaos” names three different things in this literature, and this paper reaches a different verdict on each. The first is a *location in parameter space*: a value of a tunable control parameter at which the dynamics sit between order and disorder (Langton, 1990). The second is a *regime*: a class of behaviour, in the sense in which a cellular automaton is called class IV rather than class III. The third is a *phenomenon*: frozen and moving structure coexisting in one lattice, neither absorbing the other, which is what makes such systems recognisable on sight before any measurement is taken. We close off the first inside this family, decline the second on the evidence available, and exhibit the third. What replaces the first is not a regime classifier but a causal measurement—how far

a single-cell perturbation travels—which the rest of this paper shows is governed by the currency of the rewriting layer’s access to the state it rewrites.

The parameter-space sense is closed off inside this family, by a theorem and by a scan. Our classification lands on the classical value from an unexpected direction—every fixed rule of every operator in the family is balanced, with Langton activity $\lambda = 1/2$, exactly the value the classical picture associates with the transition. A coordinate that takes one value across every fixed point cannot discriminate among them. The parity dichotomy then shows that the algebra does not settle the dynamics either: all odd-cycle operators remain genotype-alive, while every all-even cycle type contains both sustaining and collapsing members, so knowing an operator’s fixed-point structure does not tell one what it does. Searching the Meta Cellular Automaton (MetaCA) for a critical transition does not produce one (Supplement 1, Box 6): the dynamics show a broad, drifting crossover and no critical point of that sort at all.

What replaces it measures causal propagation, and is deliberately not a regime classifier. The substitute requires no parameter sweep and no choice of where to draw a region—only a perturbation and a clock. Flipping a single cell and tracking the divergence against an otherwise identical run, on a scale calibrated against elementary rules of undisputed behaviour, places the phenotype-reading construction of *Constructive control by temporal schedules* between rules 90 and 54, below rule 110 and far below rule 30, while frozen rules return exactly zero (*A Causal Measure*). That scale separates systems that propagate a perturbation from those that do not, and the separation is sharp. It does not classify regime, and we make no such claim from it: class III rules bracket its range at both ends—rule 90 at 8.00 and rule 30 at 37.50—with the class IV rules 54 and 110 between them, so a position on the axis is not a Wolfram class. Intervals on it are named below for the reference rules that bound them, and assert nothing about the systems inside them.

On that measurement, what governs propagation is the currency of the rewriting layer’s access, and no other tested coordinate. The live phenotype-to-genotype coupling is

what places the construction where it sits: cutting that single edge, with seed, tape and construction otherwise identical, roughly halves how far a perturbation reaches. The coupling is moreover a dial rather than a switch. Its gain interpolates between a genotype that reads only an ancestral phenotype and one that reads the currently acquired state, and because the ancestral field is itself a real phenotype, marginal statistics and spatial structure are matched at every setting; only currency varies. Reach follows it smoothly, while a control matched in rule mobility but blind to the phenotype does not move.

Pursued far enough, the same measurement rejects a tempting substitute. Across seven mechanisms causal reach initially appears to rise and then fall with sustained genotype diversity, but the descending observations obtain high diversity only through mutation or by holding rules still. A conservative eighth construction prescribes the genotype measure and transports it by local bijective exchanges, sustaining causal amplification across the full 32–256 rule range: exchange rate, not diversity alone, controls reach (*Diversity and conservative transport*). Across λ , propagator fraction, sustained rule diversity, mutation rate, refuge probability, niche width and rule mobility, causal reach stays low. Blend strength alone moves it (Supplement 1, Box 10). The coordinate that governs is therefore not a truth-table parameter but how much of what the genotype reads is causally current.

The same coordinate is recovered when the coupling is made a function of the lattice rather than a setting. A composition whose condition is blind to the phenotype reproduces the dial, linear in its duty cycle, so conditional composition is not by itself a new mechanism. A composition whose condition reads the phenotype exceeds that line, by margins growing with the width of the neighbourhood read. An otherwise identical gate reading a phenotype frozen at the moment of perturbation does not exceed it, and in every tested pair falls slightly below, though the two share width, spatial statistics and firing rate. Current information is thus necessary for the departure; how much of the width gradient it accounts for is not decomposed here. The coordinate of the preceding paragraph is reached a second time, from a construction sharing none of its dials.

Posed from inside the system, access becomes a limit on what can be regulated, and the third sense of the phrase is what survives that limit. Those measurements are made from outside. The corresponding question from inside is whether the construction can locate such a regime using only quantities available to it as it runs. Part III removes the last element held fixed—the rewriting operator, which becomes a property of the cell rather than of the run—and searches the resulting parameter space. The constraint that governs the dynamics governs the search as well. Causal reach is computed by forking a run and perturbing a twin; it is available to the experimenter and never to the system, and a controller cannot regulate on it. Across the same sweep, internal observables predict an intrinsic objective at held-out $R^2 = 0.73$ and a twin-run objective at 0.02. The loop is closable, and not on the measure that identified the coordinate. The objective supplied to the search fails for a second reason. It reads a fixed window and scores how structured the output looks; the behaviour that separates these configurations occurs outside that window. Its two maxima are a configuration that survives and one that freezes permanently, and it ranks the second above the first. What survives that critique is the third sense of the phrase, because it is a property of a trajectory rather than of its appearance: whether the sheet reaches a state it does not leave again before the run ends. Three configurations are separated by it. In one, live regions overrun frozen ones; in a second, the reverse happens and the lattice stops changing at a locatable step; in a third, neither outcome obtains for as long as the run can be afforded, with frozen regions still turning over after ten thousand steps. That third configuration is the phenomenon the 2015 example was recognised by. It is exhibited here, not located by any coordinate, and no self-driving search was achieved.

The scope of each claim differs, and the parts are not interchangeable. The exact results hold for the entire $8!$ core. The causal measurement covers one principal construction at one lattice width over six seeds, and no comparable causal census of either the $8!$ or 8^8 family is attempted here. The configurations of Part III sit at one point of a 40,320-member design space of rewriting operators, and no experiment varies that choice.

Part I’s scan and Part III’s boundary concern different objects on different axes— genotype aliveness under a propagator fraction, and phenotype persistence under a selection parameter—and neither bears on the other. Nothing in Part III establishes a critical point, and at the two points tested we have evidence against one. Supplement 4 reports a separate exploratory analysis of coexisting spatial domains and documents a boundary-transport statistic withdrawn after a circularity check; neither is used to support any claim here. Supplement 5 documents the search apparatus and its negative results.

The division of labour follows the argument. Part I classifies the permutation-propagators, surveys their spatial dynamics, and tests the parameter-space sense. Part II drops the restriction to a single permutation-propagator and measures causal propagation in update architectures built on the same genotype–phenotype substrate. Part III removes our hand from the coupling twice: first the condition under which an operator fires becomes a function of the lattice, then the operator itself becomes a property of the cell, and the phenomenological sense of the phrase appears there. Figure 1 maps that division and marks where the scope boundary falls.

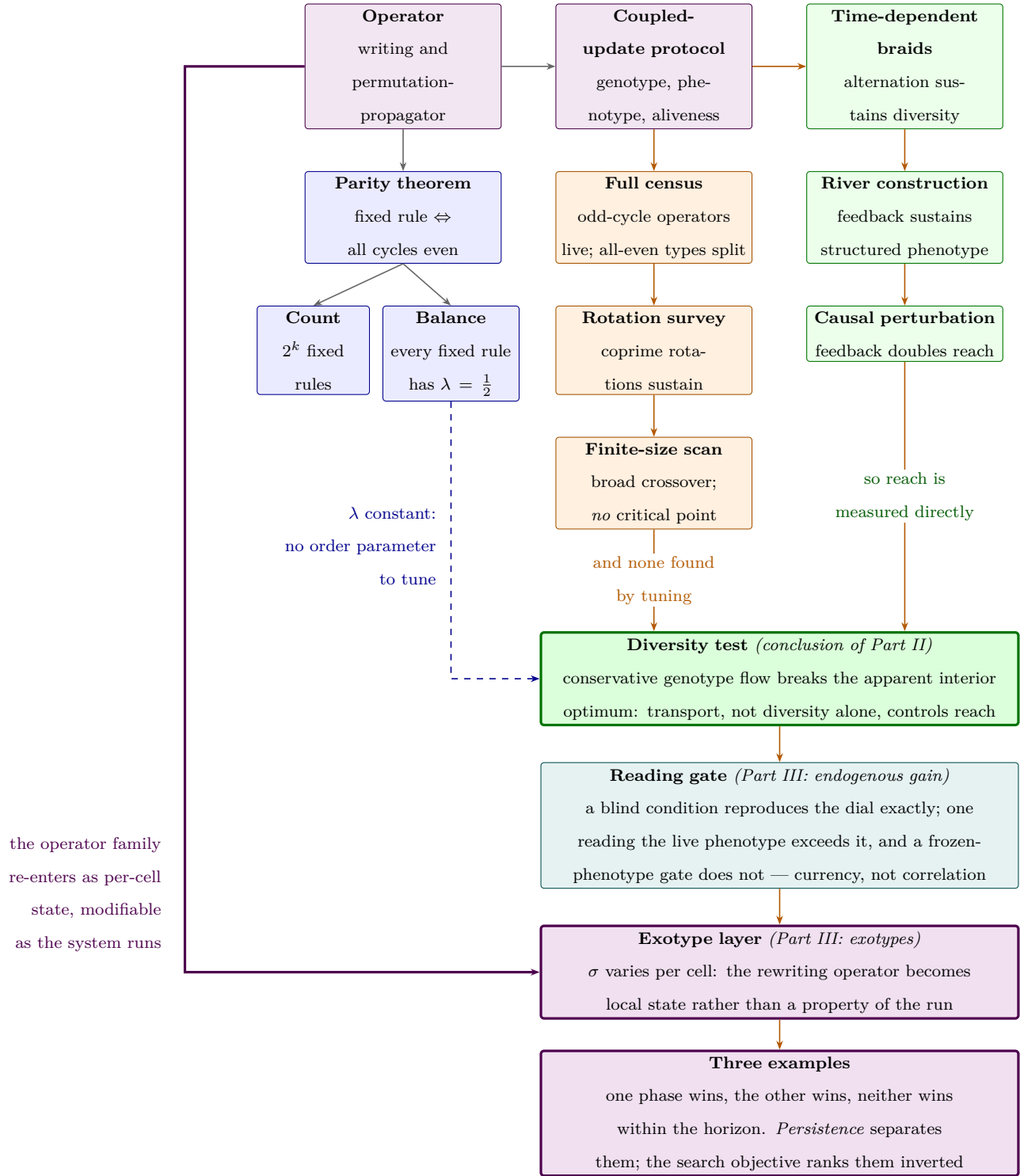


Figure 1

Argument map and scope boundary. Part I is the exact branch (blue) and the parameter-space branch (amber); $\lambda = 1/2$ closes off the classical coordinate there. Part II is the constructive branch (green), concluding in the diversity test. Part III makes the coupling endogenous twice: the firing condition becomes a function of the lattice (teal), then the operator itself becomes per-cell state (violet). The heavy violet edge closes the map — the operator family re-enters as something the system modifies as it runs.

Part I

The Permutation Core

Definitions, theorems, corollaries and worked examples are stated and proved in Supplement 2, and referred to here by number. The narrative gives each result in prose at the point where it is used, so this part can be read without turning to the supplement; the supplement carries the formal statements, the proofs, and the arithmetic worked by hand.

The 2015 Example

The construction that began this work is recreated in Figure 3. Its rule-rewriting map is not a permutation of the eight truth-table positions but an arbitrary map on them, so it lies outside the bijective core this paper classifies. What made it worth returning to is visible rather than measured: frozen and moving structure persist together across the sheet, neither absorbing the other. Locating that example within a family large enough to enumerate is what the rest of Part I does.

The Object

A cellular automaton normally evolves a field of *states* under one fixed rule. Here each cell also carries a *rule*, and local writes evolve that rule field alongside the phenotype. We display an elementary cellular automaton (ECA) rule in Wolfram order,

$$111, 110, 101, 100, 011, 010, 001, 000,$$

and index these displayed positions from left to right by $p = 0, \dots, 7$. Thus position 0 is the 111 output and most significant bit, while position 7 is the 000 output and least significant bit. Throughout, the indices of σ , w , g , and $\text{bit}[p]$ are these *displayed positions*, not the binary values of the neighbourhoods. If $n \in \{0, \dots, 7\}$ is a numerical neighbourhood value, its rule output is therefore $g(7 - n)$.

We study the operator P_σ , parameterised by a permutation $\sigma \in S_8$ of the displayed positions. Writing the displayed byte as a map $g: \{0, \dots, 7\} \rightarrow \{0, 1\}$, the operator sends g

to $P_\sigma g$ defined by

$$(P_\sigma g)(q) = \neg g(\sigma^{-1}(q)), \quad \text{equivalently} \quad \text{bit}[\sigma(p)] := \neg \text{bit}[p]. \quad (1)$$

Each output bit is one input bit, at the position σ moves it to, negated; P_σ is thus a permutation of the eight coordinates composed with a global complement. The truth table is thereby itself an eight-cell automaton whose wiring is σ and whose update is Boolean negation — an automaton inside the automaton. Equation (1) defines a family of $8! = 40,320$ operators that is finite, exhaustively enumerable, and, as we show below, exactly classified.

The legacy engine stored its table in the order 000, 001, 010, 100, 011, 101, 110, 111. Whenever a legacy writing is imported, the reproduction code conjugates it into the displayed Wolfram order above before we quote its permutation or apply names such as offset +2.¹

Worked example 1 works through one member of the family bit by bit.²

The simultaneous operator acts on one eight-bit rule; the spatial experiments place its elementary writes inside a two-layer automaton.³

The substrate—a cellular automaton whose cells hold rules rather than states—is not itself new (Corneli & Maclean, 2015); what we study is the *operator* of Equation (1), understood as an enumerated, exactly classified family. The example in that 2015 paper is a non-bijective writing (*The 2015 Example*) and therefore lies outside this family; what it shares with the family is the elementary write, not membership. We give the family here an essentially complete characterisation. The canonical symmetry group of ECA rule space has order four (reflection $\times$ complement, collapsing 256 rules to 88 classes) (Schaller &

¹ Supplement 2, Definition 1: Writings, elementary writes, and propagators. Supplement 2, Observation 1: Permutations form a small bijective core.

² Supplement 2, Worked example 1: A simultaneous propagator by hand.

³ Supplement 2, Definition 2: Genotype and phenotype. Supplement 2, Worked example 2: One site, one complete MetaCA step.

Svozil, 2025); it is not S_8 , and the negation-carrying, σ -parameterised family of Equation (1) does not appear in it. Rule-space permutation as a parameterised, enumerated family with a derived fixed-point structure does not appear in the literature we have surveyed; the nearest neighbours augment cellular automata with state-dependent feedback (Pavlic et al., 2014) rather than permuting the rule itself. Two established genres are adjacent but distinct: per-cell rules (non-uniform or ν -CA (Dennunzio et al., 2011)) and rules that change during evolution (rule-changing or programmable CA (Mori et al., 1998)). Non-uniform cellular automata are not, by themselves, the source of novelty here: when each site draws its rule from a finite set, the resulting system can be represented as a uniform cellular automaton on a product alphabet (Dennunzio et al., 2011). Our contribution instead concerns the permutation-propagator family and its fixed-point structure.

The Classification

Fixed points exist iff every cycle is even

P_σ has a fixed rule if and only if every cycle of σ has even length; since a one-cycle is odd, this already forces σ to be fixed-point-free.⁴

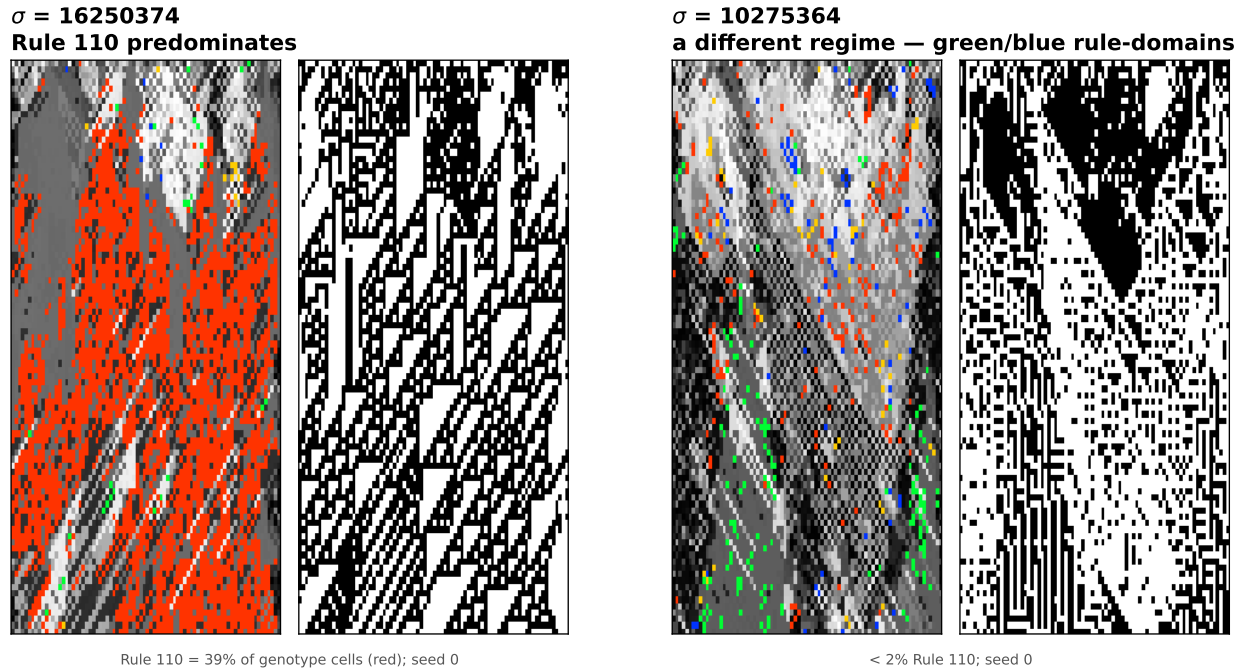
A fixed rule is a g invariant under (1), i.e. a σ -equivariant Boolean colouring with $g(\sigma k) = \neg g(k)$. Follow a cycle of length L : equivariance forces $g(k) = (\neg)^L g(k)$, and $(\neg)^L$ is the identity exactly when L is even; conversely, colour each orbit by the parity of its distance from a chosen representative. A one-cycle of σ ($L = 1$) is odd and so admits no such colouring, which is why even cycles are required throughout.

Worked example 3 shows the theorem on the offset-+2 operator. The same small calculation also makes the later 2^k count and forced $\lambda = 1/2$ balance visible before we state them generally.⁵

This is the signed-cycle fixed-point theorem of Boolean-network theory (Aracena,

⁴ Supplement 2, Theorem 1: Fixed points and even cycles.

⁵ Supplement 2, Worked example 3: Even and odd cycles.

**Figure 2**

Two operators from the family, each shown over time as genotype (256-colour, left) and phenotype (black and white, right). Under $\sigma = 16250374$ the genotype field comes to be dominated by ECA Rule 110—the red cells, 39% of the genotype at the representative seed (35% over six seeds)—and the phenotype accordingly displays Rule 110’s characteristic gliders; under $\sigma = 10275364$ a different regime of rule-domains forms (green and blue, under 2% Rule 110). The operator does not merely stir rule space: it drives it toward particular, computationally distinguished rules. Genotype colours are the legacy engine’s own.

2008) specialised to (1): our operator is a Boolean network whose signed interaction digraph is σ with every arc negative, a negation-cycle of length ℓ has sign $(-1)^\ell$, and an isolated negative cycle admits no fixed point while a positive (even) one admits exactly two. We contribute a machine-checked proof of the specialised statement in Lean 4 / Mathlib.⁶

The closed form

The number of $\sigma \in S_{2n}$ with all cycles even is $((2n-1)!!)^2$, with exponential generating function $\sum_n ((2n-1)!!)^2 x^{2n} / (2n)! = 1/\sqrt{1-x^2}$ (the “set of even cycles” construction). For S_8 this is $105^2 = 11,025$ of 40,320, or 27.34%. The count is classical: it is OEIS Foundation Inc. (2011) and appears as Stanley (1999, Problem 5.34(c)) and in Riordan (1968). We claim it only as the enumeration of our operator family, not as a new identity.

Fixed bytes number 2^k

An all-even σ with k cycles therefore has exactly 2^k fixed bytes, and any σ with an odd cycle has none.⁷ Each even cycle admits exactly two alternating colourings, chosen independently across cycles, whence 2^k ; the zero case is Theorem 1. This too is machine-checked.⁸ Summed against these weights the family is consistent: $\sum 2^k$ over all-even σ equals $(2n)! = 8! = 40,320$, the total number of fixed bytes across the family.

Every Fixed Point Is Balanced

Every fixed point has $\lambda = 1/2$

Every fixed point of every operator in the family has exactly four one-bits—equivalently, Langton’s activity parameter is pinned at $\lambda = 4/8 = 1/2$.⁹

⁶ `DarkTower/Patterns/Propagator.lean`, theorem `hasAlternatingColouring_iff_cycleType_even`; `lake build clean`, no `sorry` or `axiom`.

⁷ Supplement 2, Corollary 1: The 2^k count.

⁸ `alternatingColouring_card_eq_two_pow_cycleType_card`.

⁹ Supplement 2, Theorem 2: Every fixed point is balanced.

By Theorem 1 each cycle has even length $2m$; an alternating colouring assigns such a cycle exactly m ones; summing over cycles gives $(\sum L)/2 = 8/2 = 4$.¹⁰ Exhaustive enumeration confirms it: across all 40,320 operators and all their fixed points, every one has Hamming weight four, with zero exceptions.¹¹ The statement is machine-checked for Fin 8.¹²

We are deliberate about the standing of this result. Langton’s critical value was obtained empirically, as a statistic over rule space (Langton, 1990), and its status as an order parameter was contested: Mitchell et al. (1993) refuted Packard (1988)’s specific genetic-algorithm clustering claim, not the definition of λ , which survived as one axis among others such as Wuensche’s Z (Wuensche, 1999). There is a folklore intuition for why $1/2$ is special — binomial sampling variance $\lambda(1 - \lambda)$ is maximal there — and it is not ours; and it has been noted that λ alone does not fix a rule’s dynamical class, distinct classes coexisting at a single value (Sakai & Kanno, 2002). What Theorem 2 adds is narrower and firmer than any of these: under the complementation in (1) the value $1/2$ is *forced* as an algebraic matter, because any alternation-under-complement colouring is balanced by parity. We claim balance, exactly; any connection of that balance to dynamical behaviour we treat as interpretation, not as part of the theorem.

The cast: which rules survive, and how many

Under an all-even σ , the fixed points of P_σ —the rules it maps to themselves—number 2^k , where k is the number of cycles of σ (*The Classification*): each even cycle admits two alternating colourings, chosen independently across cycles. In the spatial runs these are the rules onto which the field is observed to settle. Every one has $\lambda = 1/2$ (Theorem 2), so the observed terminal population is a cast of *distinct* rules—distinct truth tables—sharing a *single* activity.

¹⁰ Supplement 2, Definition 3: Langton’s activity λ .

¹¹ Supplement 2, Observation 2: Balance is a truth-table property.

¹² `alternatingColouring_true_card_eq_four`: the true-cells and false-cells are put in bijection by σ , forcing each to number four.

The smallest nontrivial case of the count is a single cycle, $k = 1$: two survivors. The 2015 paper’s Figure 8 exhibited a two-rule residual of its own: a random field of tens of distinct rules dies off to an alternating population of just two. That resemblance is not an instance of this count. The writing in that example is non-bijective (*The 2015 Example*); the count above is a statement about permutations; and that writing admits no fixed rule whatever, its residual being a hunt on a single unsheddable position rather than a pair of fixed points. Two-rule residuals arise here for a different reason than they do there, and we keep the two apart.

That die-off curve—the count of distinct rules present over time—is a natural population-level companion to λ . Where λ is a scalar assigned to a single rule, this curve is a property of the whole field’s trajectory; and where λ is uninformative here—every fixed point sits at $1/2$ —the curve still varies, because it measures the population rather than the point. We use it cautiously, as an observable rather than an established order parameter, to tell the rotations whose fields stay diverse from those whose fields collapse (*Fixed Points Are Not Attractors*, Figure 6).

Diversity of the observed population and singularity of its activity thus coexist; the metric measures we tried do not separate the two (*Limits of the Tested Distance and Complexity Measures*).

A dead genotype can carry a live phenotype

The balanced shell is a definite object: exactly $\binom{8}{4} = 70$ rules carry four active outputs, and by Theorem 2 the absorbing set of every operator in the family lies wholly inside it. Collapse is therefore not merely a loss of diversity but a fall onto this particular 70-rule balanced shell at $\lambda = 1/2$.

Balance is necessary for a genotype to freeze and silent about what the frozen rule then does. Run each of the 70 as an ordinary elementary CA and they split: 32 are phenotypically live—sustained change under iteration, the additive rules 90, 105, 150 and the class-IV rule 54 among them—while the rest fall quiescent. The two halves share a

single λ ; this is, inside our family and by construction rather than by survey, the coexistence of dynamical classes at one activity value that has long been urged against λ as an order parameter (Sakai & Kanno, 2002). A field can thus die in its genotype while the surviving rule keeps its phenotype in motion.

Figure 4 is such a field, built to specification. To seat a collapse on a chosen live rule R it suffices to pair R 's four active positions with its four inactive ones as an involution: four transpositions, an all-even operator that holds R among its fixed points (Theorem 1). Whether the field actually settles onto that fixed rule is a separate, dynamical question (as *Fixed Points Are Not Attractors* stresses); here it does. Carried out for Rule 54 this yields $\sigma = [23015476]$; run from a random field it drives the genotype to near-uniformity—two rules survive—while the phenotype never settles. The field has died onto Rule 105, another live tenant of the same shell, whose additive dynamics keep the black-and-white panel full of structure. Genotypic death and phenotypic death are not the same event.

One caveat marks the open edge of this. An involution fixes not its target alone but the whole shell compatible with its pairing, and which tenant a random field actually reaches is set by the basins, not by the construction: we aimed at 54 and the dynamics preferred 105. Existence is settled—live survivors are constructible—but *steering* a field onto a named rule is not, and is the natural next question.

The Census

The parity dichotomy in the spatial dynamics

Theorem 1–2 concern fixed bytes; the spatial dynamics adds stochastic elementary writes and, in the census, neighbour blending. Figure 5 shows why fixed-point existence is not itself a dynamical verdict: without blending, all-even rotate+2 settles within the run while all-even rotate+1 remains diverse; the odd-cycle examples, which lack fixed bytes, also remain diverse. The full census below (*The full census, tabulated by class*) makes this quantitative over all 40,320 operators (Table 1); the dynamical content is the *death-rate difference* between the classes, not the class sizes.

A prospective test: reduced rotate+2

To guard against reading structure into a completed enumeration, we fixed a prediction before running the case. The reduced rotate+2 operators act as a single even cycle on a subset of positions and leave the remaining positions as one-cycles; by that leftover they are *not* all-even, so by Theorem 1 they possess no fixed point and should fall on the no-fixed-points side of the dichotomy—no rule for the field to settle onto, and correspondingly less extinction. They do (Figure 5, right): they stay diverse rather than dying off. Absence of fixed points forbids settling onto a single rule; it does not forbid short-period behaviour, and we claim only the observed persistence of diversity, not literal aperiodicity.

The full census, tabulated by class

The trend becomes an exhaustive finite-run observation once every operator is tabulated.¹³ We ran all 40,320 permutation-propagators and grouped them by cycle type (Table 1).¹⁴

Fixed Points Are Not Attractors

Possessing a fixed point (Theorem 1) is not the same as reaching one, and P_σ does not by itself move a rule toward $\lambda = 1/2$.¹⁵ Because P_σ complements every bit, $\lambda(P_\sigma g) = 1 - \lambda(g)$: the operator *reflects* activity across $1/2$, exactly preserving the distance $|\lambda - 1/2|$ rather than contracting toward it. And because P_σ is a bijection on bytes, a generic rule lies on a periodic orbit and simply cycles—its fixed points are fixed, but they are not attractors and have no basins. The only rules pinned at $\lambda = 1/2$ are the fixed points themselves. Attraction in the experiments therefore belongs to the stochastic elementary-write dynamics, with or without neighbour coupling, rather than to iteration of

¹³ Supplement 2, Definition 4: Census protocol and aliveness.

¹⁴ Supplement 1, Box 1: The full permutation census.

¹⁵ Supplement 2, Worked example 4: Reading the rotation survey.

Table 1

Genotype-alive fraction by cycle type under the census protocol, over all 40,320 permutations. Any odd cycle forbids a fixed byte (Theorem 1); empirically, every such operator is alive. The all-even types split, with collapse becoming more frequent as the number of cycles k increases.

cycle type	count	genotype-alive
any type with an odd cycle (17 types)	29,295	100.0%
[8]	5,040	82.9%
[2 6]	3,360	70.0%
[4 4]	1,260	66.7%
[2 2 4]	1,260	49.2%
[2 2 2 2]	105	25.7%

the simultaneous map P_σ .¹⁶

Constructive control by temporal schedules

Everything so far has held one operator fixed for the length of a run. Relaxing that—alternating the elementary spatial updates of two operators from step to step, rather than composing them into a single map—turns out to be constructive rather than merely destructive, and it is the first place in the paper where a schedule does something neither of its constituents can.

Braiding two operators that each collapse alone can sustain the field.¹⁷ Figure 8 shows the case: offset +2 (two 4-cycles) and offset +4 (four 2-cycles) are each genotypically dead run on their own, and alive when braided. The schedule is also a usable control rather than a switch. Mixing a sustaining operator with a collapsing one in fixed proportion over

¹⁶ Supplement 1, Box 2: The rotation survey.

¹⁷ Supplement 2, Definition 5: Temporal braid. Supplement 1, Box 3: A braid of two collapsing operators can sustain the field.

a periodic twelve-step schedule moves late-window rule diversity smoothly with the mixing fraction (Figure 9), which is what later lets us treat diversity as a dial to be turned rather than a property a run happens to have.¹⁸

Each operator negates every bit, so under composition the flips cancel in pairs: a product of two is a *pure* coordinate permutation with no negation left.¹⁹

Exploratory searches over products of two and three simultaneous propagators found no additional structural regime.²⁰ Temporal braiding is a different operation: it alternates stochastic elementary writes inside the spatially coupled dynamics rather than replacing them by one composed map.²¹

Algebraic composition is therefore closed in the sense of Proposition 1. Temporal scheduling is different: it retains each intermediate coupled state, which is why the braids of *Constructive control by temporal schedules* can produce dynamics absent from either held operator. A second constructive mechanism pairs a propagator with a step of a different kind.

Limits of the Tested Distance and Complexity Measures

Two measures were tried before the causal one and neither separated the regimes. A normalised compression distance on the rule field tracks how many distinct rules are present rather than how they are arranged, and a block-entropy estimate over the phenotype is dominated by the same quantity. Both therefore report diversity, which *Diversity and conservative transport* shows is not the coordinate that governs propagation. That failure is what motivates measuring a perturbation directly rather than summarising a field.

¹⁸ Supplement 1, Box 4: The mixing fraction is a reproducible diversity control.

¹⁹ Supplement 2, Proposition 1: Composition parity.

²⁰ Supplement 2, Definition 6: Interrupter.

²¹ Supplement 1, Box 5: The rotation split occurs with and without neighbour blending.

A Parameter-Space Test

The survey (Figure S3.1) exposes a sharp behavioural split within a family of provably fixed-point-bearing operators: only the single 8-cycles (offset $\pm 1, \pm 3$) sustain a high-diversity phase, while offset ± 2 and ± 4 collapse. A reader is entitled to ask whether that phase sits at a *critical point*—tuned between an ordered and a disordered phase, as CA complexity is often supposed to (Langton, 1990). This is the first of the two readings distinguished at the outset—an edge of chaos located in *parameter space*, at some value of a tunable control parameter—and the finite-size scan below is the test of it, the only place in this paper where that reading is put to the question. We then ask whether the sustained diversity is spatially organised into coexisting dynamical regimes.²²

Figure 10 collects the four diagnostics. The scan is over the propagator fraction q , from all blend at $q = 0$ to all propagator at $q = 1$, at lattice widths $L = 30$ to 240 with 32 seeds each; the point of varying L is that a genuine transition sharpens as the system grows while a crossover does not. Genotype activity fits a logistic in q at every size ($R^2 > 0.99$), but the crossover width does not shrink over these sizes ($w \sim L^{-0.04}$). The size-scaled run-to-run spread $L \text{Var}(a_G)$, used here as a susceptibility proxy, does peak inside the crossover and grows about 5.7-fold, but its maximum wanders—from $q = 0.75$ to 0.50 and then to the sampled boundary $q = 1$ for the two largest sizes—rather than settling. Finite-horizon collapse becomes rarer with size, and the Binder cumulant has no common crossing. Each diagnostic is individually consistent with a critical point; none of them locates one, and together they support a broad crossover instead.

The genotype field is also spatially organised. Tinting by activity resolves it into coexisting low- and high-activity domains whose boundaries trace fractal-like filamentary walls through spacetime; those walls carry elevated genotype churn, and at matched activity the two sides hold measurably different rule populations, so they are not an artefact of the tint that draws them. We report that structure in Supplement 4, together

²² Supplement 1, Box 6: No finite-size evidence of a critical point over the tested range.

with a measure of information transport along those boundaries that we built, validated against three nulls, and then withdrew when it proved to be an artefact of its own construction. Neither is a third edge, and nothing in the argument below depends on them.

Part II

Beyond the Permutation Core: Feedback and Causal Propagation

Part I's classified object and exhaustive census were the finite permutation core; its temporal braid was the boundary case, already replacing one held operator with a schedule. We now retain the same genotype–phenotype lattice and elementary rule semantics but drop the single-operator restriction. The objects in Part II are update architectures: time-dependent schedules, phenotype-conditioned writes, mutation and holding protocols, and conservative local transports. Some use writings from the 8^8 family as components, but a state-dependent architecture is not itself one fixed writing. Consequently, the results below neither classify nor measure the $8!$ family as a whole; they test how causal propagation changes when the genotype update can read the phenotype.

A Feedback-Coupled Construction

The next construction crosses that boundary by allowing the update to depend on the live phenotype. Figure 11 shows a representative run at $L = 240$: alternating phenotype domains branch, merge and reorganise for the length of the run rather than resolving to a homogeneous or short-period field.²³

A Causal Measure

A causal test also avoids a general problem with observational regions. Any statistic computed over a region drawn from the field it is computed on invites circularity, and no null defined only by moving that region can necessarily detect it. We therefore stop drawing regions: perturb one cell and ask where the effect arrives.

We run the phenotype-reading construction of *A Feedback-Coupled Construction* to

²³ Supplement 2, Definition 7: River. Supplement 1, Box 7: The river sustains structured phenotype.

$t^* = 60$, fork, flip a single phenotype bit at site x , and continue both branches to $T = 120$, recording which cells differ. The fork is exact: the construction draws one pseudo-random position per cell per step, so re-running from the same seed gives both branches an identical tape and every divergence is the causal consequence of the flip. The control is the matched feedback-off variant—same seed, tape, construction and initial state, with only the live phenotype-to-genotype edge cut. Figure 12 sweeps the flip over all 80 sites and records damage at a sequence of horizons.²⁴

Two readings of that result must be separated. A glider carries a fixed-width packet at constant velocity; a conductive medium disperses. Tracking the damage centroid distinguishes them, and the reference values must be measured rather than assumed—elementary rule 110, whose gliders support universal computation, does not give a fixed packet under a random single-site perturbation, because the perturbation creates and destroys gliders which then collide.²⁵

Figure 13 places every construction on that one axis. All points share the protocol of Supplement 1, Box 9, and the calibration reproduces rules 0, 204 and 90 exactly against it, so the reference values drawn from the elementary rules carry over unchanged. The rows split the constructions by coupling rather than by provenance, and the split is clean. Nothing whose genotype update is blind to the phenotype clears rule 90—not even the ungated transport controls, which are fully mobile and merely cannot see what they are moving through. Everything that reads the phenotype lands inside. The river appears in both rows, at 12.97 with its edge and 5.51 without, so the arrow between them is one construction moved by adding an edge rather than by changing a parameter.

The same sweep also gives the growth of the damage cone and the displacement of its centroid (Figure 14); we report those exponents descriptively and draw no inference

²⁴ Supplement 1, Box 8: Phenotype-to-genotype feedback roughly doubles causal reach.

²⁵ Supplement 1, Box 9: Causal reach on a scale calibrated against elementary rules. Supplement 1, Box 10: The order parameter is the strength of the phenotype-to-genotype coupling.

from them about gliders or particles.

The coupling as a dial

Treating that coupling as present or absent understates it. In both constructions that read the phenotype it is a quantity, and causal reach is a graded, monotone function of it.²⁶

For conservative transport the gain is the phenotype-gated swap rate; reach crosses rule 90 between rates 0.10 and 0.20 and saturates past 0.75. For the river it is the *currency* of the phenotype bits the genotype reads: each cell at each step reads the live phenotype with probability γ and a frozen reference otherwise, so γ is the fraction of the genotype's view that is causally current. Because the frozen field is itself a real phenotype, marginals and spatial structure are matched at every γ and only causal currency varies; because $\gamma = 1$ reproduces the live river exactly, the dial is anchored against a measurement made independently of it.

Two consequences bear on the reading of *A Causal Measure*. First, the $\gamma = 0$ endpoint is 1.28, not the 5.51 of the ablated control: that control freezes the phenotype the genotype reads but still presents a field which, in the perturbed branch, contains the flip. It cuts the *currency* of the read, not the read. The doubling reported for it is therefore the smaller part of a coupling effect that spans an order of magnitude.

Second, the two dials have opposite curvature (Figure 15). Transport is concave and buys most of its reach early; the river is convex, with 45% of its span in the last eighth of γ , so one stale read in eight costs 40% of the reach. Monotonicity in the coupling is thus common to both constructions while its shape is not, and we would not generalise the shape from two cases.

²⁶ Supplement 1, Box 11: Coupling is a dial, not a switch, and reach is monotone along it.

Diversity and conservative transport

Langton's λ is silent on this family: the classification puts *every* fixed rule of *every* operator at $\lambda = 1/2$, so it cannot distinguish an operator whose field stays diverse from one that collapses. It cannot be rescued by computing it on the realized dynamics either.

Across the nine configurations of Figure S3.2 the empirical neighbourhood distribution is close to uniform—normalised visitation entropy 0.946 to 0.999, all eight neighbourhoods visited in every run—so λ read off the realized transitions with uniform weight (0.42 to 0.52) and the same quantity weighted by visitation (0.40 to 0.51) differ by 0.035 on average, and both sit near $1/2$ wherever the nominal value does. Nor is the silence peculiar to this family: of the elementary rules calibrating the scale above, rules 30, 54, 90, 150, 204 and 232 all have $\lambda = 1/2$ exactly, spanning frozen, ordered, complex and chaotic. What the balance theorem contributes is not a new failure of λ but an exact and provable instance of a known one. The sustained distinct-rule count can, and the causal measure above lets us ask the stronger question: does causal reach depend on diversity itself, or on the mechanism used to sustain it?

Our initial survey varied sustained diversity by seven mechanisms that share little: the bare operator family; braided writings; the fork time after seeding the genotype with a randomised *full* rule population; a continuous blend strength; random rule mutation; asynchronous refuges that hold a cell's rule exactly with some probability; and spatially phase-shifted braid niches. A limiting case anchors the top of the range: a genotype held fixed at the full population has maximal diversity and no rewriting at all. The coordinate throughout is diversity *sustained through the response window*, not diversity at the instant of the fork—a field seeded with all 256 rules holds them for one step and collapses during the very window the damage is measured in.²⁷

Figure 16 shows why the apparent optimum was an artefact. The original

²⁷ Supplement 1, Box 12: Sustained diversity does not determine causal reach; conservative transport breaks the apparent interior optimum.

configurations trace a rise and fall against sustained diversity, but every point on the descending limb obtains its diversity either by replacement mutation or by holding rules still, so the descent measures those two mechanisms rather than diversity. The conservative runs, which hold the rule histogram exactly invariant while varying transport, separate into flat bands — one per transport rate — that span the whole diversity axis. Reading across a band shows the effect of diversity at fixed transport and finds almost none; reading between bands shows the effect of transport and finds an order of magnitude.

Two cautions bound this result. First, the conservative intervention is a local transposition, rather than the one-bit, one-site intervention of the original survey. Normalising by initial damage puts their amplification on a common scale, but the conservative curves are a falsification control, not a claim that their absolute heights rank all mechanisms. Second, phenotype-gated bijective exchange is one deliberately constructed transport law, tested at one lattice width and one response horizon. It establishes existence: fields can remain mobile and propagate damage at effective diversities well above 135 without mutation. It does not establish that transport rate is a universal order parameter.

The defensible conclusion is narrower and cleaner. Distinct-rule diversity describes population persistence in the Part I operator experiments, but it does not govern causal reach across the Part II architectures. The latter depends on how rewriting or transport is organised. Identifying a coordinate that captures that organisation across mechanisms is work for another paper; the classification of *The Classification* guarantees only that λ cannot discriminate among fixed points in the permutation core.

What this scale does and does not license needs saying plainly. It separates systems that propagate a perturbation from those that do not, and that separation is sharp: frozen rules return exactly zero. It is *not* a regime coordinate. Wolfram class III rules bracket the range at both ends—rule 90 at 8.00 and rule 30 at 37.50—with the class IV rules 54 and 110 between them, so a position on this axis does not by itself classify a system as ordered, complex or chaotic, and we make no such classification. What we claim is narrower and

causal: the live phenotype-to-genotype edge roughly doubles how far a single-cell perturbation reaches, on a measure that involves no region-drawing and therefore cannot fail in the way the withdrawn measure of the Supplement 4 analysis failed. It concerns the phenotype-coupled construction, not the bare operators; no comparable measurement of the family is attempted here.

What the coupling supplies

One measurement bounds how the convexity of *The coupling as a dial* should be read. Asking, per cell and step, whether the live phenotype context selects a *different* rule than a frozen context would: it does in 73.1% of cell-steps (and in 76.8% against no context), over four seeds at $L = 80$ for 120 steps. The construction's first-match runs over four candidates, so an uninformative re-selection would differ in about 75% of cases: at 73.1% a frozen context carries almost no information about the live answer. The read is determinative rather than a tiebreak, which is what a convex gain curve should lead one to expect, and it means there is no common case that a phenotype-blind rule could encode in its place. We draw no stronger conclusion from this than that: whether *any* blind construction could match the river is a question about a search we have not run, and Supplement 1, Box 8 reports only that sixteen hand-chosen ones do not.

Part III

Endogenous Coupling and Exotypes

In Part II the gain of the phenotype-to-genotype loop is a parameter we set. This part asks what happens when it is not. It does so twice, at two different levels of the construction. First the *condition* under which a rewriting operator fires becomes a function of the lattice rather than a dial we turn. Then the *operator itself* becomes a property of the cell rather than of the run. The first half studies an update architecture of the Part II kind whose

firing condition is set by the lattice; the second draws its per-cell rewriting operators from the family Part I classifies and lets the lattice choose among them.

An Endogenous Gain

A *conditional composition* applies one operator where a local predicate holds and another where it does not. Throughout this section the first is conservative transport at unit rate, which sits high on the calibration of *A Causal Measure* at 25.38, and the second is holding, which sits near its floor at 1.21; the predicate decides, cell by cell and step by step, which applies, and its *firing fraction* f is measured from the run rather than prescribed. Two disciplines inherited from the gain dial of *The coupling as a dial* make the comparison meaningful: the transport stream advances at every opportunity whether or not the gate fires, so a phenotype-reading predicate cannot desynchronise the branches of a damage fork; and where the predicate consumes randomness, those draws come from a separate stream re-seeded identically in both branches.

A gate firing at a prescribed rate, blind to the phenotype, is described across six rates spanning $f = 0.10$ to 0.94 by a line through the holding value,

$$\text{reach}(f) = 1.21 + 22.42 f, \tag{2}$$

with $R^2 = 0.993$. A blind conditional composition is therefore *exactly* a gain dial, its duty cycle playing the role γ plays in *The coupling as a dial*. The construction adds nothing on its own; what matters is what the condition reads.

Letting it read the phenotype changes that. Writing agree k/m for the condition that at least k cells of the centred circular m -cell phenotype neighbourhood agree with the centre, thirteen such gates spanning widths $m \in \{3, 5, 7, 9\}$ all exceed Equation (2) at their own measured firing fraction, several by a wide margin. The widest permissive gate, agree 5/9, reaches 38.23 while firing on only 63% of opportunities — above unit-rate transport's own 25.38. Firing a chaotic operator less often, in the right places, spreads damage further than firing it always. What predicts the size of the departure is the *width*

of the neighbourhood read rather than how hard the gate is to satisfy, though strictness is nearly collinear with f and the two are not cleanly separable here.

That admits a deflationary reading, and the control that rules it out is the one that matters here. A gate reading m cells has neighbouring decisions sharing $m - 1$ inputs, so width *is* the spatial correlation length of the firing pattern, and correlated firing of a transport operator composes adjacent swaps into longer-range movement. A *frozen* gate applies the identical predicate to the phenotype as it stood at t^* : same width, same spatial statistics, a real phenotype field, and therefore the same correlation structure — but no ability to track where a perturbation has since spread. Across seven pairs matched on (m, k) the live gate exceeds its frozen counterpart in every case, by 14.5 cells on average, and every frozen departure from the blind line is *negative*, between -0.04 and -4.18 . The comparison is conservative in the direction that matters: the frozen gate fires *more* often than its live counterpart in six of the seven pairs, and still reaches less.

The gate is therefore a targeting mechanism, and targeting works only on current information. An endogenous gain does not escape the causal currency of what the genotype reads; it re-derives it. A composition whose condition is blind reproduces the dial exactly, and one whose condition reads the phenotype exceeds it precisely to the extent that what it reads is current.

From Gate to Operator

The gate chooses *when* an operator fires. It does not choose *which*. Throughout the preceding section the pair of constituents is fixed, and the predicate selects between them; the permutation σ supplying the rewriting is a property of the run. The remainder of this part removes that restriction. Each cell carries its own propagator, drawn from a small named vocabulary and transmitted between neighbours, so the operator acting at a site becomes part of the local state.

The Question This Part Does Not Answer

The motivating question is easy to state and we have not answered it. The Part II architectures and the endogenous gate are placed on a scale by an instrument we hold: damage reach, computed by forking the run and perturbing a twin. The question is whether the construction can dispense with that instrument—whether a MetaCA carrying an exotype layer can find its own operating point using only quantities available to it at runtime. Damage reach cannot be that quantity, since no system can fork itself; and the coordinate it measures is causal propagation rather than dynamical regime, so it would not by itself say where the system should arrive.

We built an apparatus to test this and it did not settle the question. Supplement 5 documents it in full, because its negative results are specific enough to be reusable, and because one of them bears directly on the question: a runtime-computable pair of observables tracks an intrinsic objective well ($R^2 = 0.73$ held out) and damage reach not at all ($R^2 = 0.02$). What the supplement does not contain is a system that searched and found. The examples reported below were located by that apparatus together with our own reading of what it returned, and we are not able to separate the two contributions.

We therefore make a weaker claim, and state it plainly so that it is not mistaken for the stronger one. **A self-driving search, had it worked, would have been expected to return configurations with robust intermediate behaviour. We report such configurations.** They are reported here as objects with measured properties, not as the output of an autonomous procedure, and they establish that the targets of such a search exist in this substrate rather than that the substrate can find them.

The Layer

An *exotype* is a per-cell assignment of a propagator. The vocabulary is a set of named permutations σ ; the grid holds one name per site; and the names are transmitted between neighbouring sites by a local policy. Nothing in the vocabulary is new to this paper—each name denotes a member of the family Equation (1) defines—but the

assignment is new, because σ has until now been a property of a run rather than of a cell.

The transition is one elementary write. At a site whose genotype is the eight-bit rule g , with σ the propagator the site currently carries, a coordinate $k \in \{0, \dots, 7\}$ is drawn uniformly at each step and

$$g'(\sigma(k)) = \neg g(k), \quad (3)$$

with g unchanged elsewhere. Comparing this with *The Object*: the simultaneous operator P_σ applies Equation (3) at all eight coordinates at once, and the spatial experiments of Part I place its elementary writes inside the two-layer automaton. The exotype layer is that same elementary write, drawn one coordinate at a time, with σ read from the cell rather than from the run.

Three properties of this transition are worth stating because the informal vocabulary the project has used around it—exotype as *culture*, transmissible but not heritable—obscures all three.

First, **the exotype edits the rule; it does not change how the rule is applied.** σ never touches the propagation of phenotype state. It selects a write address inside the genotype. The word *propagator* is unhelpful on exactly this point, and we retain it only because Part I has established it for the operator family.

Second, **when σ is the identity the transition is a point mutation**—flip a uniformly chosen bit of the rule. For every σ other than the identity the read address and the write address differ, so the operation transfers inverted information from one entry of the truth table to another. An exotype is therefore best described as a locally varying, structured mutation operator, whose bias is which bit of the rule is rewritten from which.

Third, **it fires at every applied site at every step.** It is the ordinary path, not a rare perturbation. In the configuration measured here it is also the dominant one, though we report the split without the per-event protocol needed to audit it. *3% from the policy blend.*

We also drop the phrase *transmissible but not heritable*. The exotype grid persists

across steps and is transmitted between sites; whatever term is preferred, that is inheritance of an acquired modifier in the sense that a Baldwin-style argument requires, and declining the word does not change the causal structure. What the layer does is better conveyed by Equation (3) than by any of these labels.

The Vocabulary Is Twelve of Forty Thousand

The implemented vocabulary has twelve members. The family Equation (1) defines has $8! = 40,320$. The layer therefore instantiates 0.030% of the permutation core, and if the family is widened to the arbitrary maps the 2015 implementation actually had, $12/8^8$ is 0.00007%.

The twelve are drawn from the family Part I classifies and organised along the coordinates that classification supplies, so Part I's theorem audits them exactly: it predicts, without any dynamical measurement, how many rules each member cannot alter, and the prediction agrees in all fourteen implemented cases including the two controls that are defined but not selectable. That agreement is an audit of the vocabulary and not a premise of anything below; no persistence result in this part depends on it.

The twelve were chosen by hand, and the reasons were not recorded. They span both cycle parities and both ends of the fixed-point count, which is what makes the layer's behaviour attributable to the exotype mechanism rather than to a degenerate choice of operator, but the selection was not designed to test any hypothesis about the family.

What Varies, and What Does Not

The boundary on everything this part claims is set by the same table. Of the original four propagators, *why those four* has never been justified; they were hand-picked, and the reason is not recorded. The eight additions were made for a documented but narrow reason: the selection dynamics had collapsed onto **collapser** because the scoring cache iterated only the four declared names, so the objective could not select the others. Widening the vocabulary repaired a defect in the search, and was not an attempt to cover the family.

Consequently every dynamical result reported below is measured at one point of a 40,320-member design space, chosen by hand for unrecorded reasons, and no experiment we have run varies σ across that space. Nothing here is known to generalise across propagators. This is not a caveat to be added at the end; it is a boundary on what the part is entitled to claim, and it is the most consequential limitation of the work.

It also identifies the extension with the clearest warrant. The vocabulary is finite and small because it is a hand-written map from names to permutations; nothing in Equation (3) requires that. Allowing σ to range over the family—sampled, or enumerated within a cycle type, or varied along the cycle-parity and fixed-point axes Part I supplies—would turn a fixed point in the design space into a coordinate, and would make Part I’s classification available as an experimental variable rather than only as a prediction about twelve cases. We have not done this, and we note it as an extension rather than as work in progress.

Three Examples

Supplement 5 records a search over two parameters of the selection rule: the policy precision β and the epistemic weight κ . The search did not become autonomous, and its objective turns out to be the wrong target. What it did produce is a map, and on that map three configurations are worth reporting as objects in their own right. They are what a working search would have been expected to return.

All three are measured the same way and by a criterion the search never used. Write a site as *frozen* at time t if its phenotype has been unchanged for the preceding fifteen steps, and *live* otherwise. A run is *absorbing* if it reaches a configuration it does not leave again for the remainder of the run. These are properties of a single trajectory, requiring no twin run and no external reference; the horizon is the only free choice, and we report it with every number.

The ridge from below

At low β the lattice is live and stays live. At $\beta = 2$, $\kappa = 0$ the frozen fraction settles at 0.040 and remains there through 3000 steps, on both seeds; the same holds at $\beta = 2$, $\kappa = 0.1$ and at $\beta = 4$ for $\kappa \in \{0, 0.1, 0.2\}$. No run in this group reaches an absorbing state.

The mechanism is not that frozen regions fail to form. They form and are then eaten. Over the first three measurement bands at $\beta = 2$, $\kappa = 0$ the number of frozen domains holds at ten while their mean width collapses from 9.0 cells to 3.8 and then to zero. A constant count with falling width is boundary retreat: every domain loses territory from both edges at once. Interior dissolution would look different — holes opening inside frozen regions would *raise* the domain count by fragmentation — and the count never rises. Live territory invades frozen territory at every boundary simultaneously, and there is nothing left at the end.

The ridge from above

At high β and low κ the same contest runs the other way, and the losing side puts up a fight worth describing. At $\beta = 32$, $\kappa = 0.1$ the lattice carries a dense branching structure through a field of static stripes, fragmenting the frozen field as it goes: the frozen domain count *rises* from fifteen to twenty-two over the second measurement band, which is live territory cutting frozen territory into pieces. It then loses. The pieces merge, the count falls to two domains of width 124, and at $t = 1046$ the configuration stops changing and does not change again before the run ends. Every subsequent row is byte-identical; the final pattern has density 0.500, so this is a frozen pattern rather than a blank sheet.

The second seed absorbs at $t = 460$, and $\beta = 64$ absorbs at $t = 881$ and $t = 537$. The outcome is seed-robust and the timing is not: absorption time varies by more than a factor of two at fixed parameters, so differences of that size between neighbouring β carry no information.

Two facts about this pair matter more than either alone. First, they are the same kind of object: both carry coherent propagating structure and sustained genotype diversity,

and they differ in which phase wins. Second, the search objective ranks them the wrong way round — 0.625 for the arm that dies against 0.422 for the arm that lives. Across the seven configurations run to 3000 steps on two seeds the separation is perfect and inverted: every configuration that reaches an absorbing state scored higher than every configuration that does not.

The example in the valley

Between the two arms the objective falls to its minimum, and it is there that the most interesting configuration sits. Bisecting β between the arms at $\kappa = 0.1$ locates the crossing between them: $\beta = 12, 14$ and 16 all absorb, at $t = 1917, 2604$ and 1722 ; $\beta = 8$ and $\beta = 10$ do not.

At $\beta = 8, \kappa = 0.1$, one realisation sustains an intermediate state for as long as we have run it. Across three windows spanning 10,000 steps the frozen fraction reads 0.071, 0.048 and 0.073 — neither collapsing to the live-arm value of 0.04 nor climbing toward one — with 391 distinct frozen regions still forming and dissolving at $t = 9500$ and no absorbing state at any point. The frozen phase is fine-grained and in continuous turnover: as spacetime connected components the regions have a median lifetime of 24 steps and a median width of 4 cells, and the largest reaches an area of only 650, so no region coarsens into a permanent continent within the observed horizon.

This configuration is the phenomenon in the third sense distinguished at the outset—frozen and moving structure coexisting, neither absorbing the other—and so the one such a search would have been built to find. It is also the one the objective ranks eighth of thirty-five, at 0.453, below both arms of its own ridge.

What these examples do and do not establish

The honest description of this region is not that neither phase wins. It is that **the parameters do not determine which phase wins**. The second seed at $\beta = 8$ loses its frozen phase entirely, reading 0.103, 0.000 and 0.001 across the same three windows. At $\beta = 10$ the two seeds reach *opposite* outcomes: one sheds its frozen phase to 0.141, the

other freezes completely and absorbs at $t = 6346$. Widening the lattice to 1000 cells changes the answer again, and since that run also extended the horizon we cannot attribute the change to width alone.

Realisation-to-realisation variance dominating a parameter's effect is what a critical region looks like, and we report it as that rather than as a regime. What the three examples establish is that this substrate contains configurations of all three kinds — one phase winning, the other winning, and neither winning within the horizon we can afford — and that they are separated by a bisectable boundary in β . What they do not establish is a critical point, which would require the intermediate behaviour to be a property of the parameters rather than of the draw, and we have evidence against that at the two points tested.

One methodological point generalises beyond these three examples and is the reason the section is measured by persistence rather than by any of the alternatives we tried. Every configuration reported here was present in the thirty-five-cell search of Supplement 5 and was scored there over a 250-step window. At that horizon the example in the valley is unremarkable and the arm that dies looks best. The behaviour that separates them — absorption at $t = 1046$, turnover sustained past $t = 9500$ — occurs entirely outside the window in which the objective was computed. An instrument validated on 250-step reference automata can be measuring a transient in a system whose transients run four times longer.

Discussion

Limits

The Part I theorems are exact for the Boolean permutation core; its dynamical claims are narrower and empirical. The census uses three seeds per operator, the rotation

survey two, and the finite-size scan 32 per parameter cell, all under the stated radius-one spatial protocol. These finite-run results do not turn fixed-point balance into a claim about attraction. The substrate itself is not new (*The Object*), and λ is one heuristic among several (Wuensche, 1999); what is exact is only that the fixed-point structure of Equation (1) forces every fixed rule in the core to have $\lambda = 1/2$.

Part II has different limits because it has a different object. Its central causal comparison concerns one feedback-coupled construction rather than either writing family, one lattice width, and six seeds. The descriptive exponents are fitted over a range where damage width reaches only a tenth of the lattice, so finite-size effects are not excluded. The architecture survey and conservative-transport control compare mechanisms rather than exhaustively sampling 8^8 writings, and their fixed sample sizes are stated with each result. They identify feedback and transport as controls in the tested constructions, not as universal order parameters. Illustrative diagrams use pinned representative seeds. All reported computations are reproducible from the accompanying package.

Supplement 4 carries the separate spatial-domain analysis and the information measure withdrawn after its mask and measure proved circular. Its claims and failures do not enter either Part I's classification or Part II's causal comparison.

The Part III results are narrower again. The endogenous-gate measurements use one pair of constituents, one predicate family and sixteen seeds per row; the frozen control's firing fractions differ from their live counterparts by up to 0.04, and the width gradient is not decomposed—a blind gate with matched correlation length would separate current information from firing geometry and has not been run. The exotype configurations sit at one point of a 40,320-member design space of rewriting operators, which no experiment varies, and use two seeds per parameter point. Those seeds disagree at two of them, so the region is reported without a critical point and with evidence against one where tested. All persistence claims are bounded by the horizons actually run, and the width-1000 comparison changed lattice width and duration together, so it is not attributable.

Conclusion

The paper has studied four related objects, and its conclusions must keep them separate: permutation-propagators, update architectures, conditional compositions, and per-cell exotype systems. Part I concerns the finite permutation core. Its static results are exact: a permutation-propagator possesses fixed rules exactly when all cycles are even; an all-even permutation with k cycles has 2^k fixed rules; and every such rule is balanced, with $\lambda = 1/2$. Its dynamical results are finite-run observations over that core. The exhaustive census shows that all odd-cycle operators remain genotype-alive under the stated protocol while every all-even cycle type contains both sustaining and collapsing members. The rotation survey and finite-size scan sharpen the distinction: fixed-point existence does not determine attraction, and the tested propagator-fraction scan gives a broad crossover rather than evidence of a critical point.

Part II retains the two-layer lattice but changes the object from an individual writing to an update architecture. The river construction composes a permutation-propagator with a phenotype-reading step; the wider comparison adds temporal schedules, mutation, holding, niches, and conservative local transport. These are not a census of the 8^8 writings, and the results do not retroactively characterise the $8!$ core. They establish a narrower causal claim. Against a matched feedback-off control, phenotype-to-genotype feedback roughly doubles the reach of a single-cell perturbation. When the strength of that feedback is varied, reach changes monotonically, while a control matched in rule mobility but blind to the phenotype remains in the low-reach band. Conservative transport further shows that sustained genotype diversity is not itself the governing coordinate: high diversity can coexist with propagation, and transport changes reach far more than diversity does.

The connection between the parts is argumentative rather than theorem-to-theorem. Part I proves that the classical λ coordinate cannot discriminate among fixed points in the permutation core and finds no critical point in the tested parameter scan. Part II asks

what a causal measurement can reveal once the architecture itself is allowed to change. Its answer is feedback coupling for the constructions tested here, not a theorem about all permutations or all writings.

This admits a reading the substrate makes unusually clean. The gain γ interpolates between a genotype that reads only an ancestral phenotype and one that reads the currently acquired state, with the frozen field a real phenotype so that marginal statistics and spatial structure are matched at every setting; only causal currency varies. What the dial varies is thus how legible acquired state is to the heritable layer—the Lamarckian question, posed as a continuum rather than a dichotomy. Its discriminating power is unusual in this family: across λ , propagator fraction, sustained rule diversity, mutation rate, refuge probability, niche width and rule mobility, causal reach stays below rule 90’s reference value, and blend strength alone passes it. The substrate carries no fitness, so this concerns what the heritable layer can see and not what is selected for.

Part III changes the object twice more. It first moves from an architecture whose gain we set to one that computes its own, and then from a fixed rewriting operator to one the lattice carries per cell. Its two halves are measured differently. The endogenous-gate results are finite-run causal measurements over a chosen family of predicates, not a characterisation of conditional compositions in general; the exotype results are persistence measurements on single trajectories, which is what the search’s own failure made necessary. What they establish is a negative and a positive that belong together. A composition whose condition is blind to the phenotype reproduces the Part II dial exactly, linear in its duty cycle—so conditional composition is not by itself a new mechanism. A composition whose condition reads the phenotype exceeds that dial, by margins that grow with the width of the neighbourhood read; but an otherwise identical gate reading a phenotype frozen at t^* does not exceed it at all, and in every tested pair falls slightly below. Width is the spatial correlation length of the firing pattern, so this is the comparison that matters: correlated firing alone buys nothing, and what the reading gate exploits is that what it

reads is current. The coordinate is therefore the same one Part II identifies, recovered from a construction that shares none of Part II's dials. That convergence is the strongest connection the paper draws between its parts, and it is still a convergence of measurements rather than a theorem. Supplement 4 records the separate spatial-domain analysis and the withdrawn boundary statistic; neither is a premise of this two-part argument.

Part III is narrower still, and narrow in two places. Its gate measurements cover one pair of constituents—unit-rate transport against holding—and one predicate family, agreement within a centred circular neighbourhood. Thirteen conditioned gates at four widths support the width gradient; seven matched pairs support the frozen control. Each row is sixteen seeds at one lattice width, so the between-seed spread is real and reported rather than smoothed. Two limits deserve naming. The frozen gate's firing fraction does not match its live counterpart exactly, differing by up to 0.04; the discrepancy runs in the direction that weakens the result, since the frozen gate mostly fires more often and still reaches less, but it is not zero. And while the frozen control shows that current information is *necessary* for the departure, it does not decompose the width gradient: how much of that gradient is currency and how much is the geometry of correlated firing is not resolved here. A blind gate with matched correlation length, rather than matched rate alone, would separate them and has not been run.

Its exotype measurements are narrower again. The vocabulary instantiates twelve of the 40,320 permutations Part I classifies, the original four were hand-picked for unrecorded reasons, and no experiment varies σ across that space, so nothing measured there is known to generalise across propagators. The three examples are single configurations at one lattice width, reported with the seeds that disagree with them: at the two points where the phases are closest to balanced, two seeds reach opposite outcomes, and we therefore report a region in which the parameters do not determine the winner rather than a critical point.

Future work. The two extensions the gain result suggested have each moved, and neither is closed. Making the gain endogenous is done: a predicate reading the lattice sets

it, and reach responds to whether what the predicate reads is current. Letting the system set its own operating point was attempted and failed, and Supplement 5 records how. An earlier controller adjusted a parameter from an internal error signal and failed on leverage and stability; the later apparatus scored a surface and moved over it, but never closed a loop, because no runtime-visible target survived validation. The remaining task is therefore not to show that adjustment is possible in principle, but to find a target a system can compute about itself that stays valid over the horizons on which persistence is decided.

The second extension is unchanged and remains the more interesting. As it stands the substrate instantiates the Lamarckian arm alone: acquired state writes into the heritable layer, and both the dial of Part II and the gates of Part III vary the strength or the aim of that write-back. The Baldwin effect requires something this substrate lacks—a selective filter, so that plasticity alters what selection sees and the heritable layer follows without direct transcription. Adding selection over a population of these architectures would supply that ingredient, and both arms could then be compared externally on causal reach. A regulator operating from inside such a population could not use that measure, for the reason Part III gives, and would need a persistence-like criterion instead; Supplement 5 sets out the conditions such an experiment should satisfy before it is built.

Data and code availability. Every numerical result in this paper is regenerated from source rather than deposited: the generators are seeded, so a single documented command rebuilds the fields underlying the tables, figures, and reported summary statistics from an empty data directory, and the analysis scripts consume no input that command does not produce. That command was run twice from an empty data directory and all 307 generated artefacts were byte-identical across the two runs; the 298 of them that also exist as published artefacts match those as well. The code and those commands are publicly available at <https://github.com/tothedarktowercame/mmca-clj> (commit 7e04a6de6afe35de041b0ac4bb52a5cea25fe8d4)—a standalone, dependency-light Clojure port of the MetaCA dynamics whose **README** gives the complete reproduction command

and commands for individual figure groups. The classification and balance theorems are formally verified in Lean 4 with Mathlib at <https://github.com/tothedarktowercame/mathlib4>, branch `darktower` (commit `bc34d8cec27216aae1d3cd511a9950ab1bce3575`), in `DarkTower/Patterns/Propagator.lean`: the cycle-parity criterion as `hasAlternatingColouring_iff_cycleType_even` and the $\lambda = 1/2$ balance as `alternatingColouring_true_card_eq_four`. The same file also contains executable regressions for both hand calculations, including the displayed-position versus numerical-neighbourhood indexing contrast.

Generative-AI disclosure. Anthropic Claude (Opus 4.8, Opus 5, and Fable 5) and OpenAI Codex (version 5.6-sol) coding agents (Anthropic, 2026a, 2026b; OpenAI, 2026) assisted in an iterative, human-supervised workflow with manuscript editing, software implementation, test construction, and adversarial checks. Exchanges with Claude Fable 5 contributed to the initial identification of the operator family reported here. The author reviewed the resulting prose and code, reran the reported computations and formal checks, and accepts full responsibility for the manuscript. Product information was accessed in July 2026; per-source access dates are given in the bibliography.

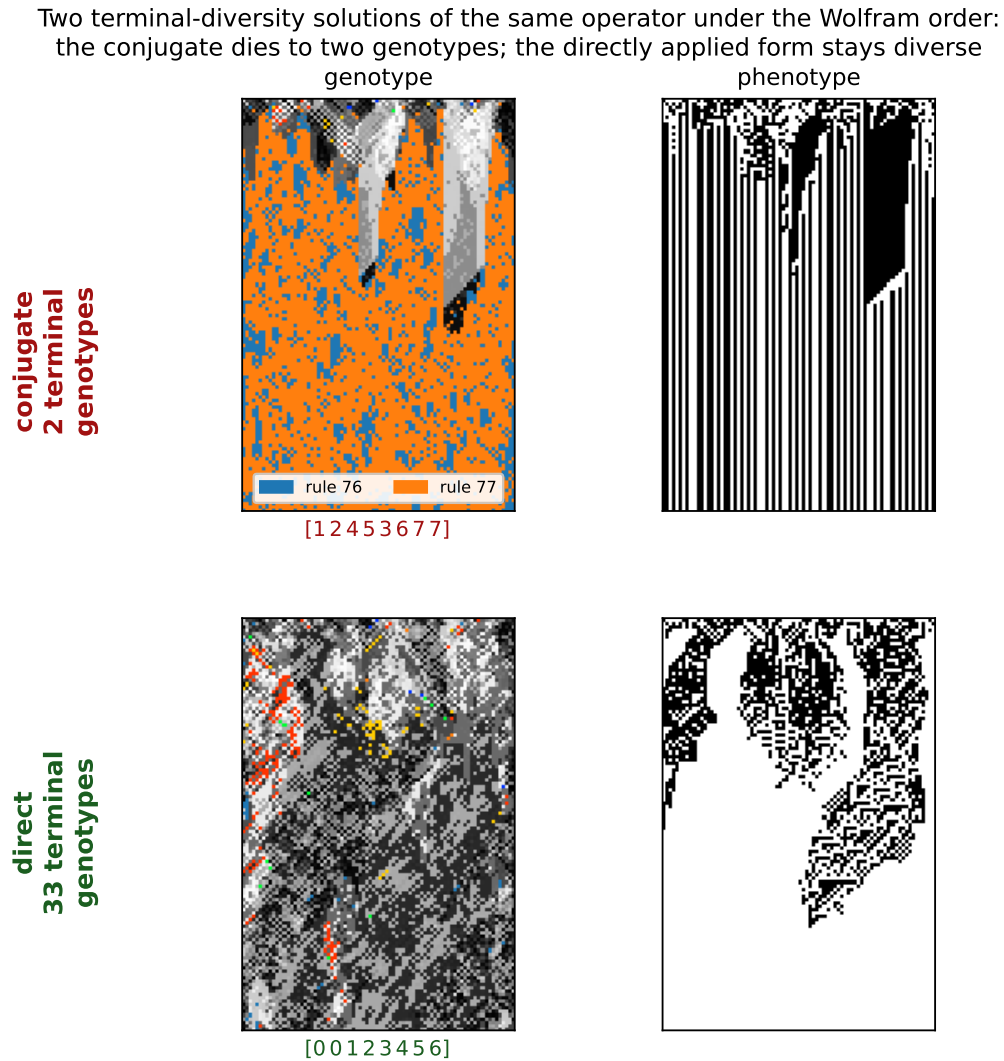
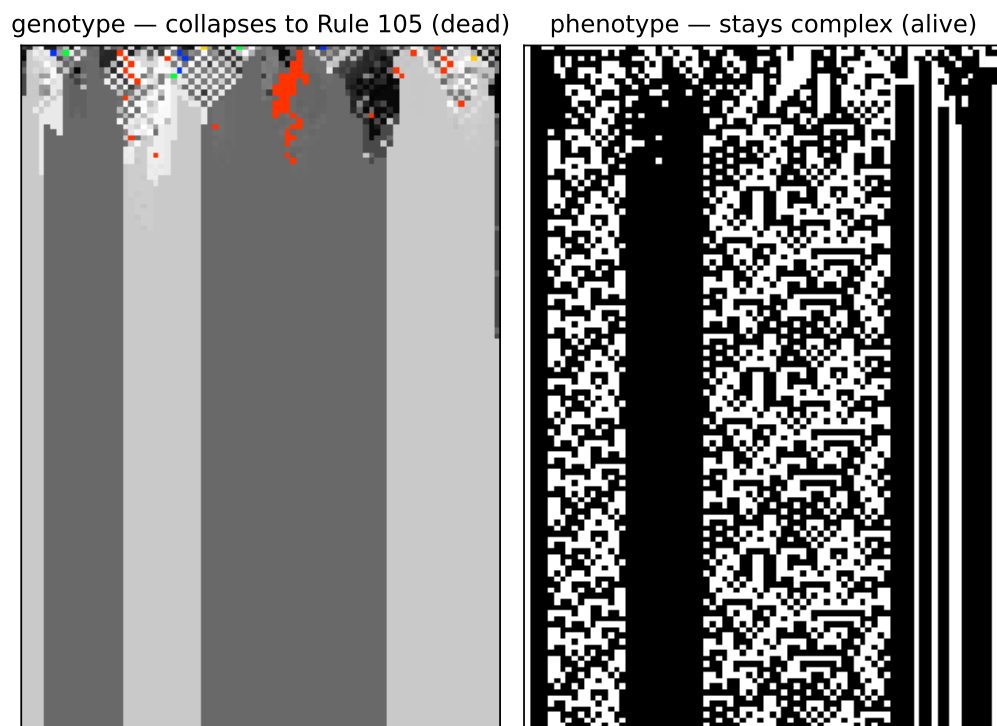


Figure 3

Two terminal-diversity solutions of one historical operator. The historical write [00123456] has two forms under the standard Wolfram order. Its conjugate [12453677] (top) reproduces the two-rule behaviour: the genotype collapses to exactly two terminal rules—76 and 77, adjacent, differing in a single bit—regardless of seed, while the phenotype stays regular. Applied directly (bottom), the same operator keeps a diverse genotype (~30 terminal rules). Two ends of genotypic aliveness from one historical write.

**Figure 4**

A genotypically dead, phenotypically live field. The all-even operator $\sigma = [23015476]$, built to fix the balanced shell containing Rule 105, collapses the genotype (left, 256-colour) to near-uniformity within a few dozen rows—two rules survive—yet the phenotype (right) never settles: the field has died onto Rule 105, an additive $\lambda = 1/2$ rule whose dynamics keep it complex. Seed 1, $W = 80$, $T = 120$.

CYCLE PARITY DICHOTOMY: a Boolean fixed point exists iff every cycle is even

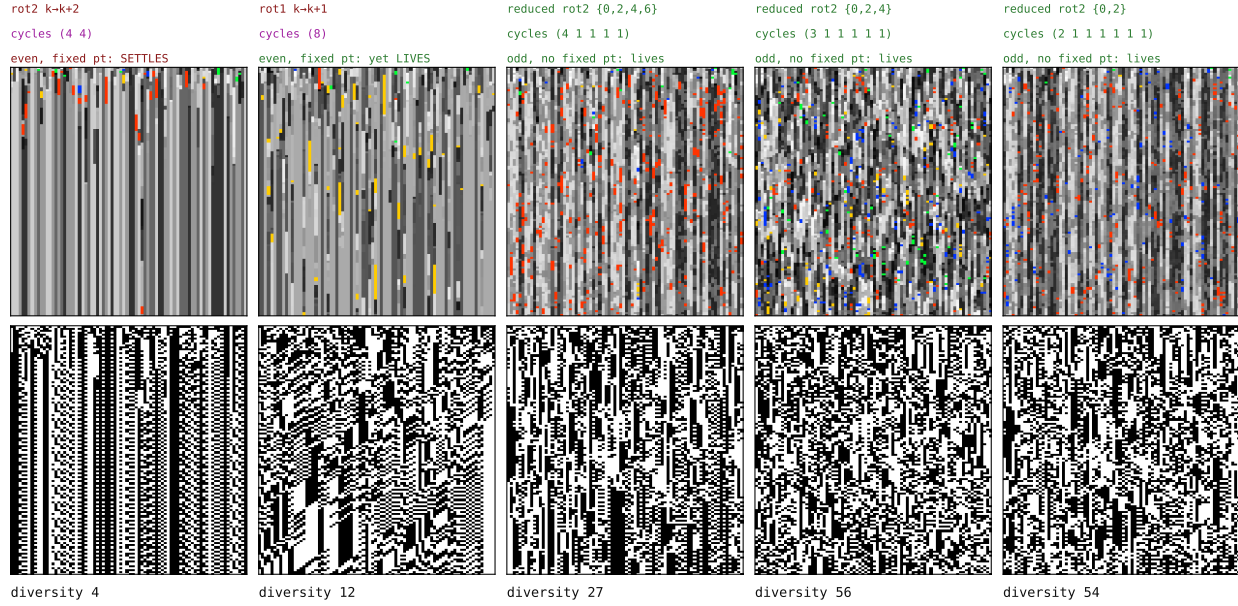


Figure 5

The parity dichotomy, by example (the census of all 40,320 permutations is summarised in the text). Five operators as genotype (top) and phenotype (bottom) over time. Left: both all-even operators possess fixed bytes (Theorem 1), but only rotate+2 settles within the run, reaching its $2^k = 2^2$ fixed rules; the single-8-cycle rotate+1 remains active at diversity 12. Right: the odd-cycle operators (reduced rotate+2, acting on a subset so the remaining positions are odd one-cycles) possess no fixed byte and stay diverse in these runs. All five use stochastic elementary writes without neighbour blending. The figure therefore illustrates the theorem's fixed-byte classification while also showing that all-even parity makes settling possible, not inevitable on a finite horizon.

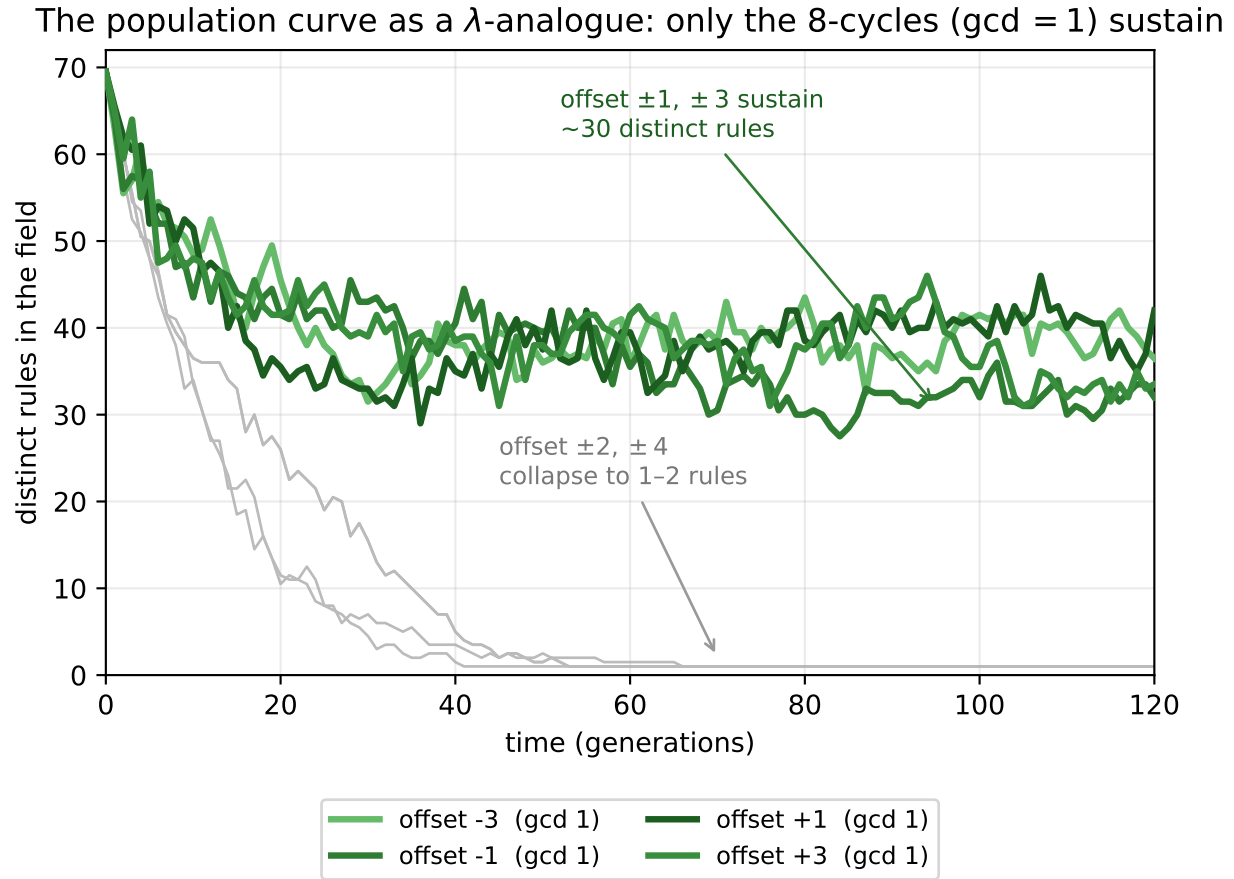


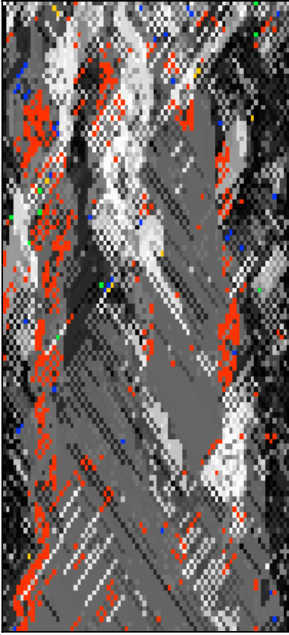
Figure 6

A population observable: distinct rules in the field over time, for the wrap rotations (two seeds each). All begin at ≈ 70 ; the four 8-cycles, $\gcd(\text{offset}, 8) = 1$ (offset ± 1 and ± 3 , green), keep a large diverse population, while offset ± 2 and ± 4 (grey) collapse to one or two rules. This makes quantitative, on the runs shown, the difference visible by eye in Figure S3.1. Figure 5 gives the corresponding without-blend comparison for offsets $+1$ and $+2$: the former remains active at lower diversity over that run, while the latter settles onto its four fixed rules.

Same cycle type (two 4-cycles), opposite fate: cycle structure does not determine aliveness

[6 7 0 2 1 4 3 5] — **two 4-cycles**

sustains (live / live)
genotype

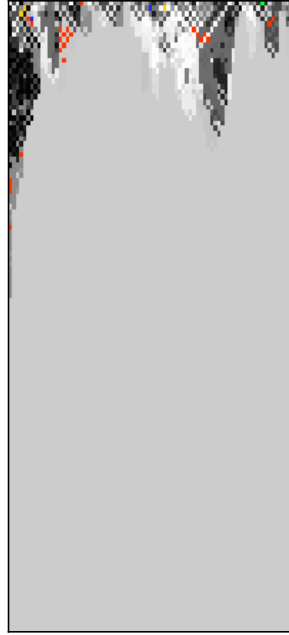


phenotype



[2 3 4 5 6 7 0 1] = **offset +2** — **two 4-cycles**

collapses (dead / dead)
genotype



phenotype



Figure 7

***Cycle structure does not determine aliveness.** Two operators of identical cycle type—both two 4-cycles—with opposite fate. Left, [6 7 0 2 1 4 3 5] sustains a live genotype and phenotype; it is the standard-Wolfram-order form of the historical offset+2 operator (its conjugate). Right, the Wolfram-order rotation offset+2 = [2 3 4 5 6 7 0 1] has the same two-4-cycle structure yet collapses to a dead genotype and a frozen phenotype. Same cycle type, opposite outcome: the specific permutation, not its cycle type alone, decides. A full census of the 40320 operators confirms the two-4-cycle class splits between sustaining and collapsing.*

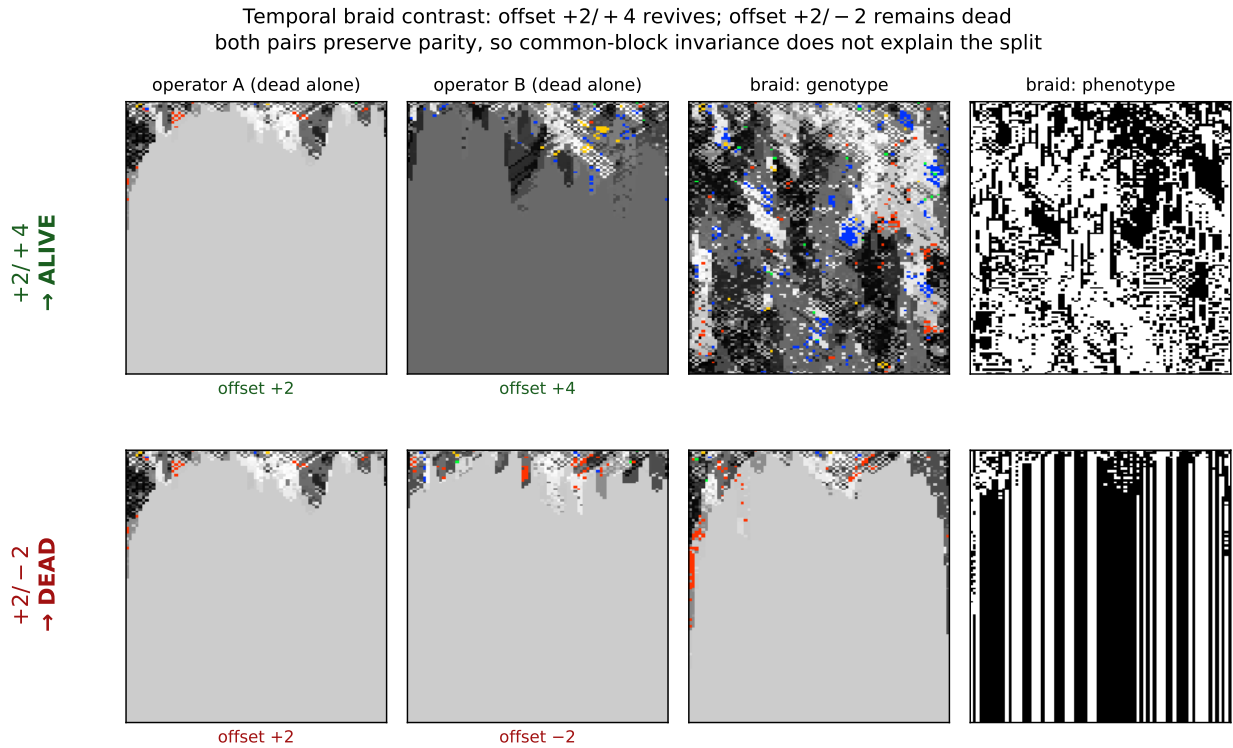


Figure 8

Temporal braiding can sustain aliveness. Two operators that each collapse alone are braided by alternating their elementary spatial updates. Top: offset+2 (two 4-cycles) and offset+4 (four 2-cycles) are each genotypically dead, yet their braid is alive (genotype change-rate 0.76, vs. 0 for either alone). Bottom: offset+2 braided with offset−2 stays dead. Both tested pairs preserve parity, so the contrast is not explained by the existence of that common invariant partition. The result concerns time-dependent coupled dynamics, not composition of the simultaneous maps P_σ .

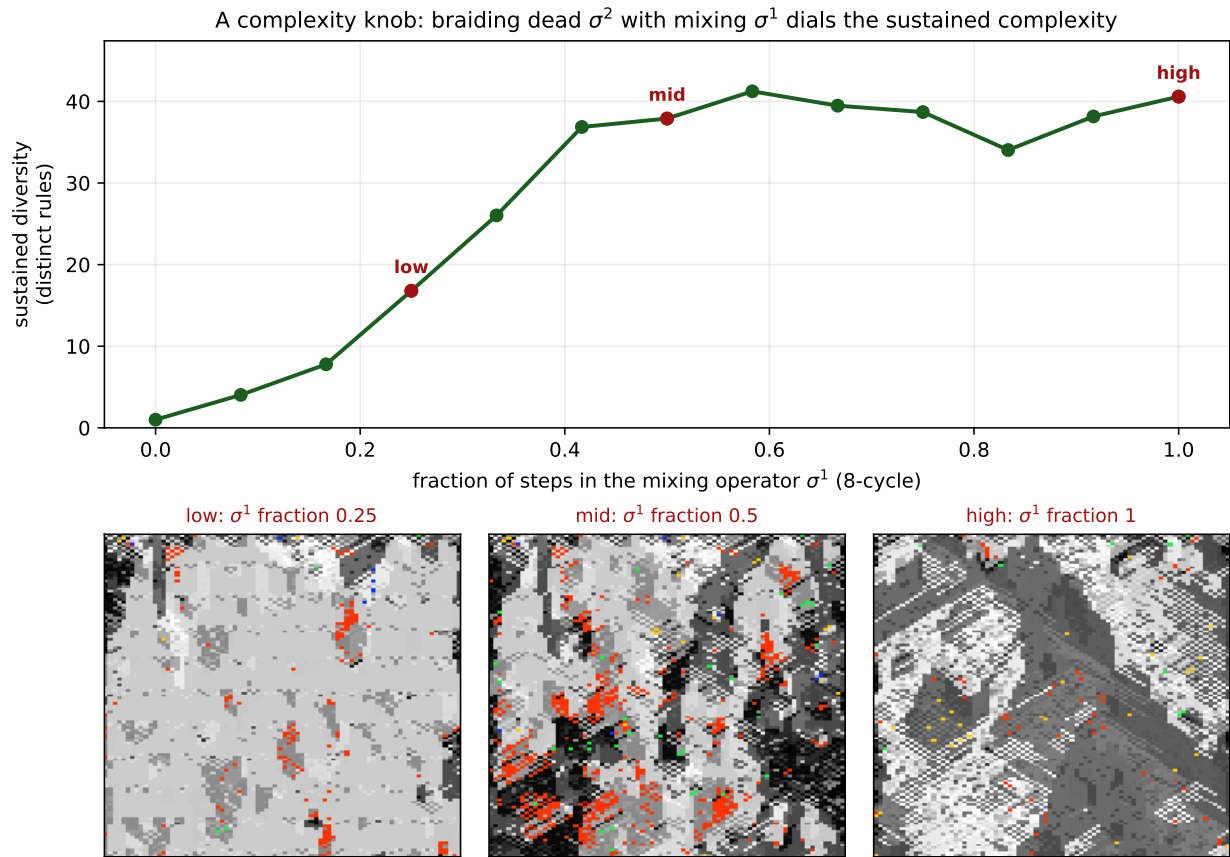


Figure 9

A diversity-control schedule. Periodic 12-step schedules mix the sustaining offset+1 operator σ^1 with the collapsing offset+2 operator σ^2 . Mean late-window distinct-rule diversity (five seeds) rises from 1.0 with no σ^1 steps to a high-diversity plateau once its fraction reaches roughly 0.4. The three fields show fractions 0.25, 0.50, and 1.0 at the same representative seed.

offset+1 feedforward scan across system size L (row width in cells, 30–240; 32 seeds): a smooth crossover with no critical point, and a susceptibility that peaks in the band but drifts and barely grows—metastability, not criticality (bottom: one realization across q , $L = 240$)

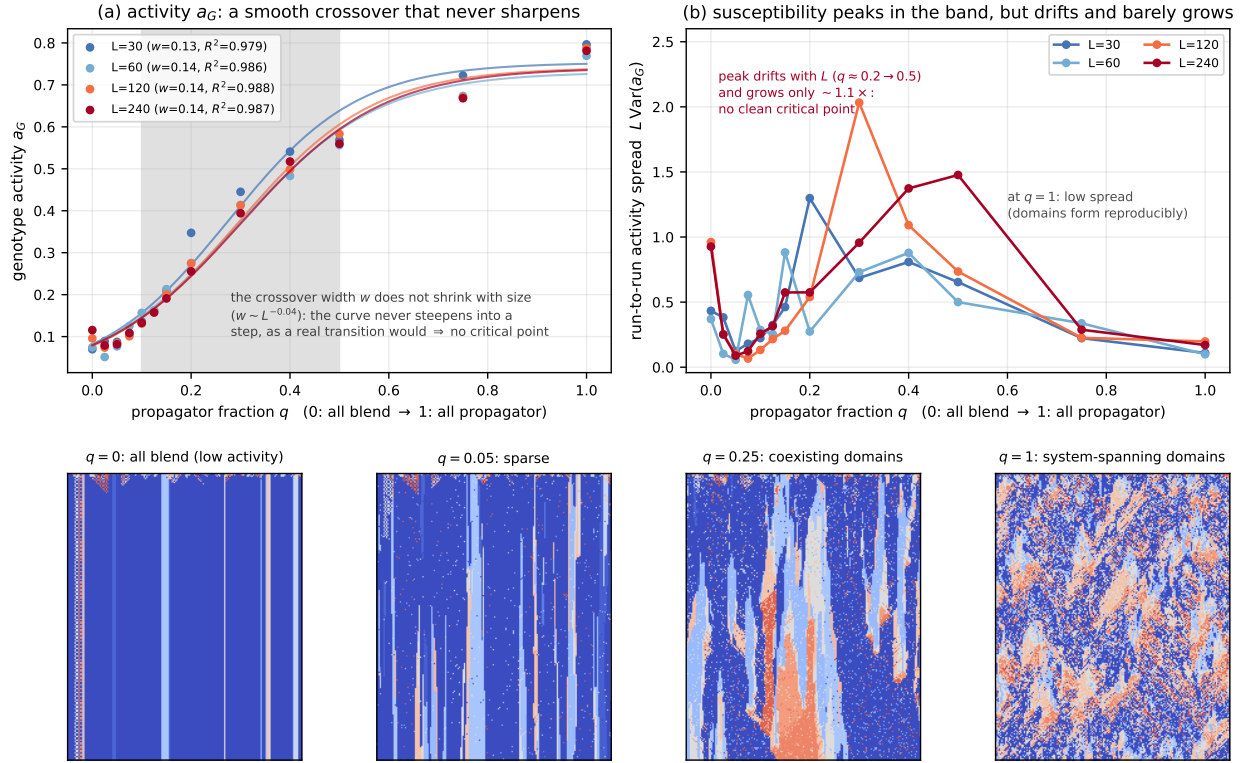


Figure 10

Finite-size scan across the propagator fraction. Offset +1 feedforward,

$L = 30\text{--}240$, 32 seeds per size. (a) genotype activity with logistic fits; (b) size-scaled spread as a susceptibility proxy; (c) finite-horizon collapse frequency; (d) the Binder cumulant.

Bottom: one paired realisation at $L = 240$ tinted by isolated-rule activity.

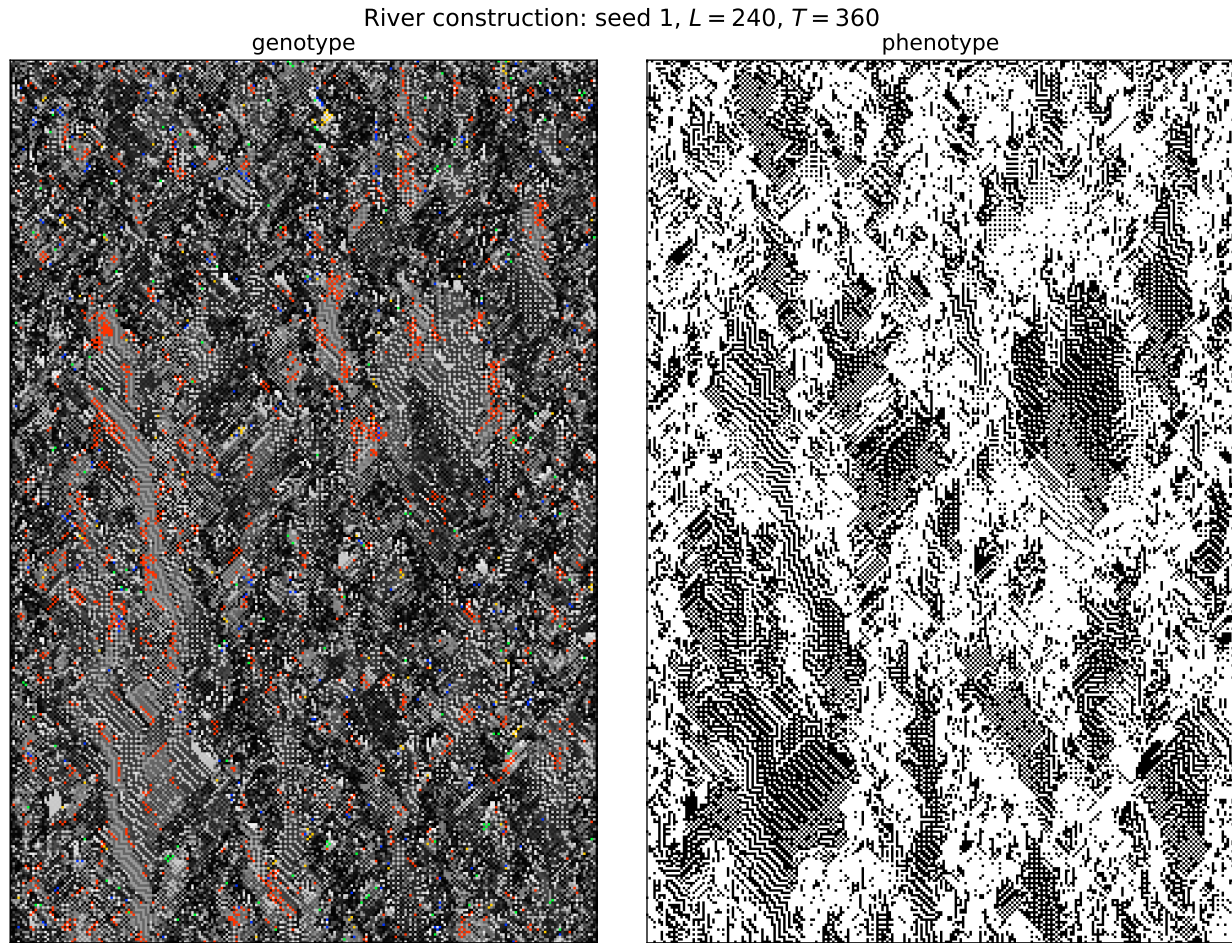


Figure 11

A larger river spacetime plate. One pinned representative run (seed 1) at $L = 240$, $T = 360$, with genotype at left and phenotype at right. The offset-+2 propagator is composed with a phenotype-reading, four-candidate, first-match template construction (fallback: the all-zero rule). Alternating phenotype domains branch, merge, and reorganise throughout the run rather than resolving to a homogeneous or short-period field. Genotype is coloured by rule as in Figures 2–Figure S3.1: greyscale by byte value, with well-known ECAs highlighted and red denoting Rule 110. Where the bare rotations collapse or cycle (Figure S3.1), this composition sustains structured phenotype for the length of the observed run.

A single phenotype-bit flip spreads farther in the river than in the feedback-cut control
(same seed, same RNG tape, same construction; single-seed sweep over all 80 sites, $L=80$)

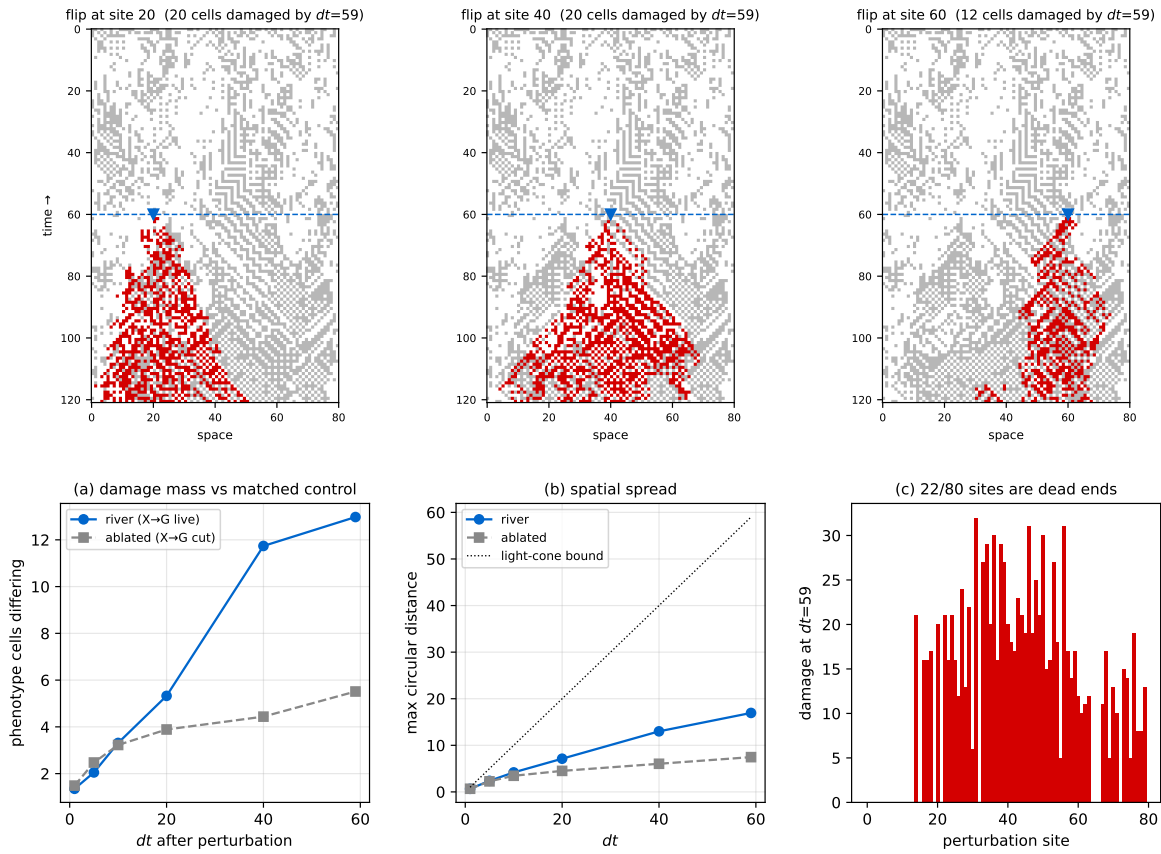


Figure 12

Damage from a single phenotype flip. Live construction against its matched control, $L = 80$, $t^ = 60$, swept over all 80 sites; mass and spread at a sequence of horizons dt .*

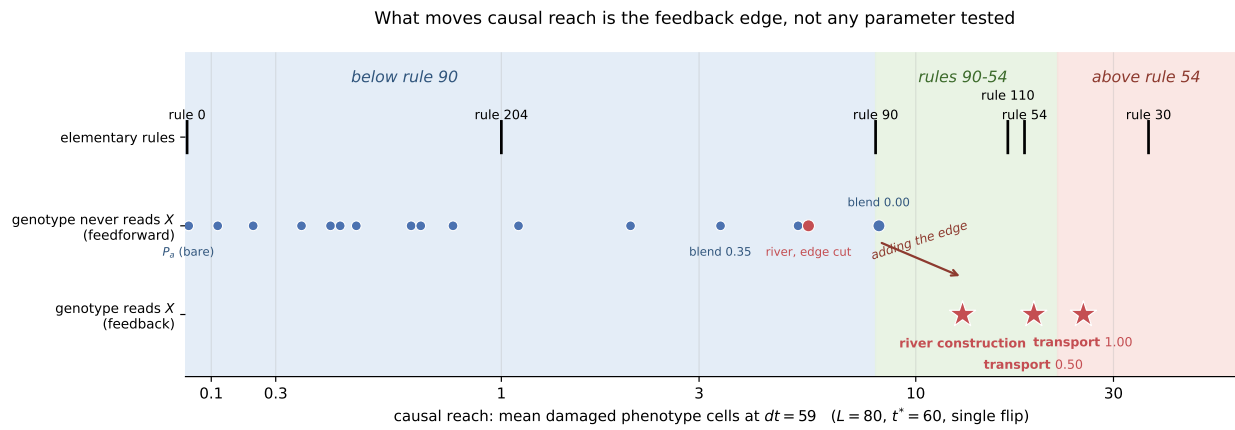
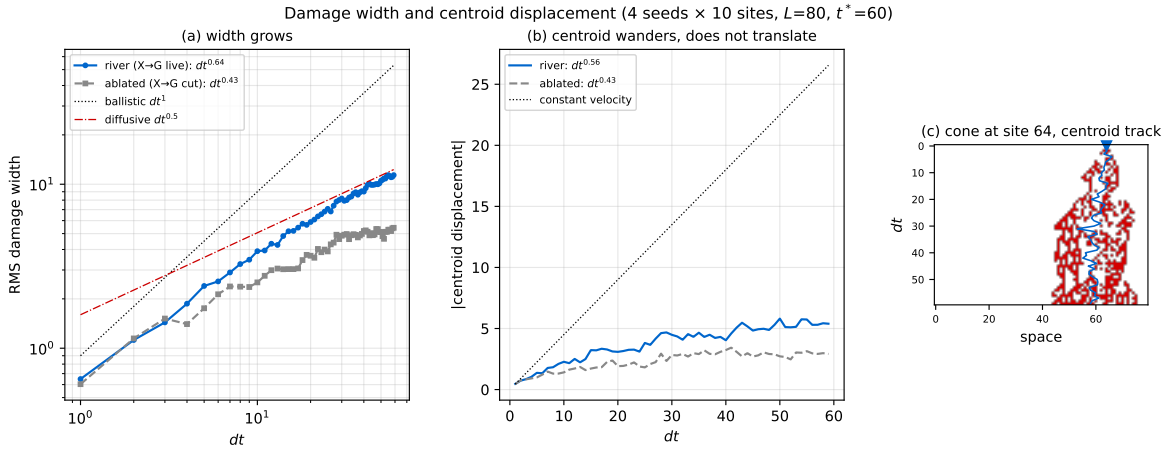
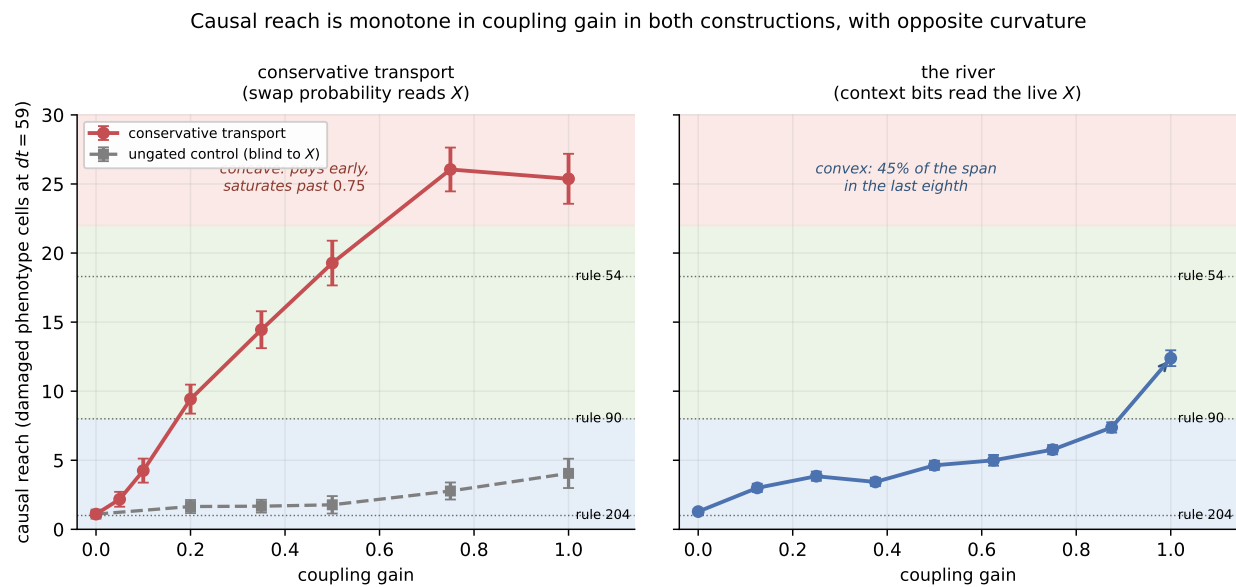


Figure 13

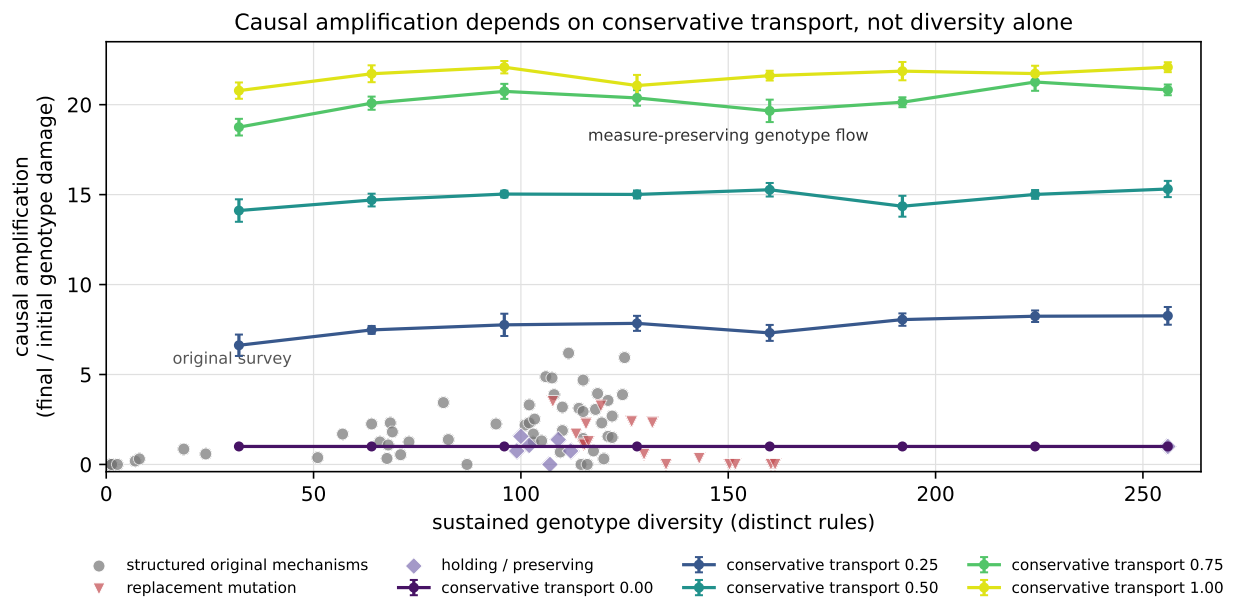
Every construction on one scale. Causal reach under the protocol of Supplement 1, Box 9. The shaded intervals are named for the elementary rules that bound them and assert no Wolfram class: rule 90 is class III and sits at the first boundary. Rows divide constructions by whether the genotype update reads the phenotype.

**Figure 14**

Damage width and centroid displacement. (a) damage width against dt , log-log, with ballistic and diffusive references; (b) centroid displacement; (c) a single cone with the centroid track overlaid. The packet disperses while its centroid stays near the perturbation site. We do not read this as excluding particle-like transport: under the same protocol rule 110, whose gliders support universal computation, gives $dt^{0.68}$ and $dt^{0.73}$, so a random single-site perturbation does not produce a fixed-width translating packet even where gliders certainly exist.

**Figure 15**

***Two couplings, dialled.** Causal reach against coupling gain for conservative transport (left, with an ungated control at matched mean swap probability) and for the river (right). Error bars are standard errors.*

**Figure 16**

Causal reach against sustained genotype diversity. The original 76 configurations, grouped by mechanism, with the conservative-transport runs overlaid as one band per transport rate.

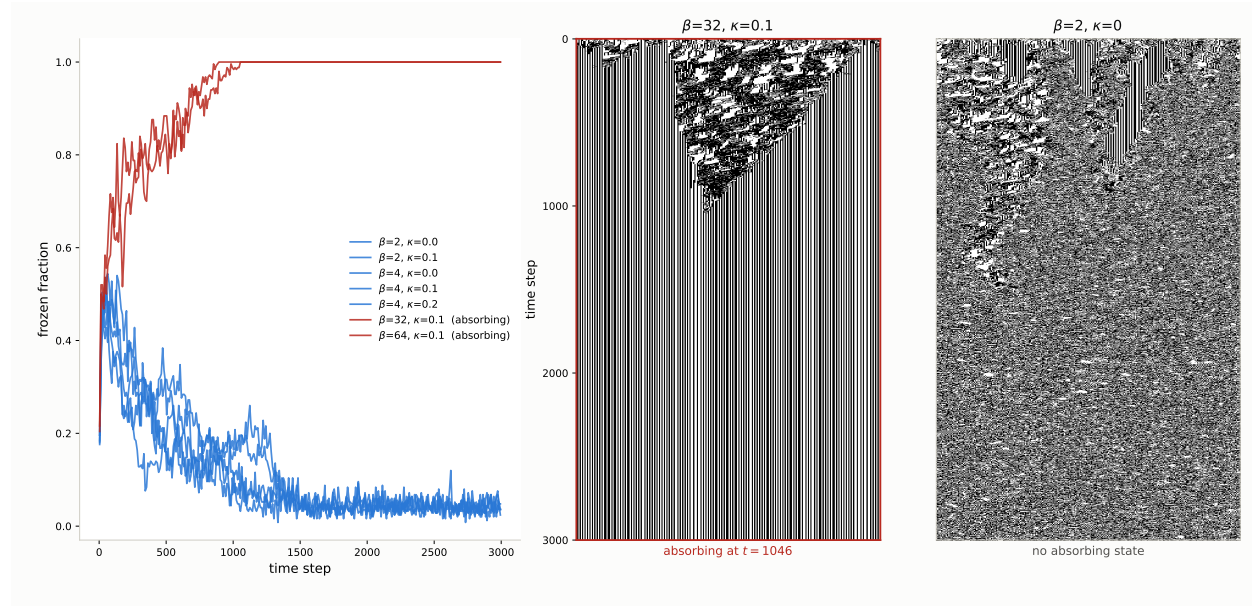
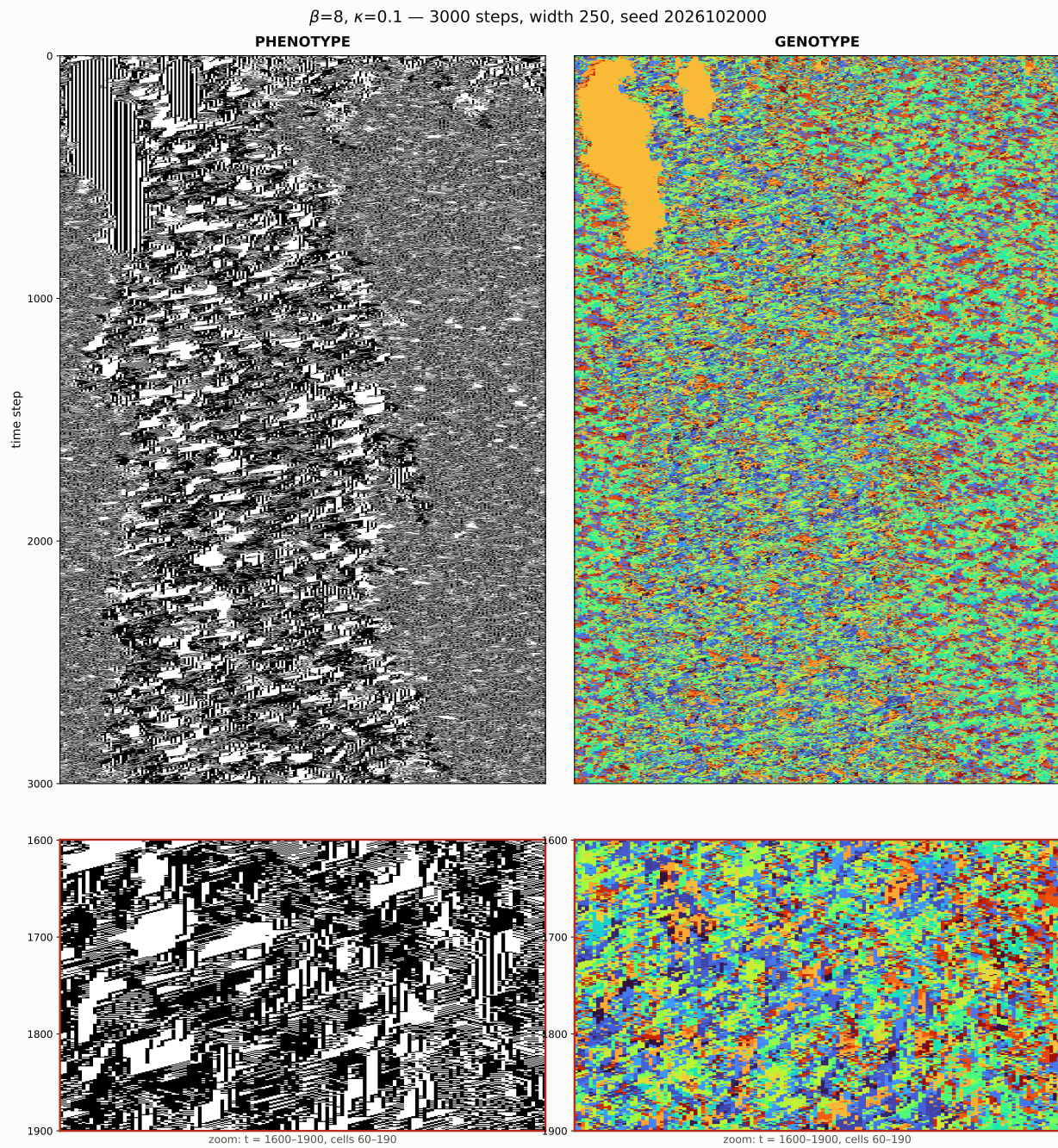


Figure 17

The two arms of the ridge. Left: frozen fraction against time for the seven configurations run to 3000 steps, with the two that reach an absorbing state drawn in red. The arms separate within the first few hundred steps and do not cross again. Centre and right: the two extremes as spacetime diagrams — $\beta = 32, \kappa = 0.1$, which fragments the frozen field for some five hundred steps and is then squeezed out, against $\beta = 2, \kappa = 0$, which erodes the frozen field to nothing and continues. The same contest, opposite winners.

**Figure 18**

$\beta = 8, \kappa = 0.1$, both layers. Coherent diagonal propagating structure runs the full sheet after the initial transient. The genotype carries one monoculture region, gone by $t \approx 500$, and thereafter remains diverse (mean row diversity 0.306, churn 0.405 per step, all 256 sigils present). The lower row zooms to the scale of the frozen regions, median 4 cells by 24 steps.

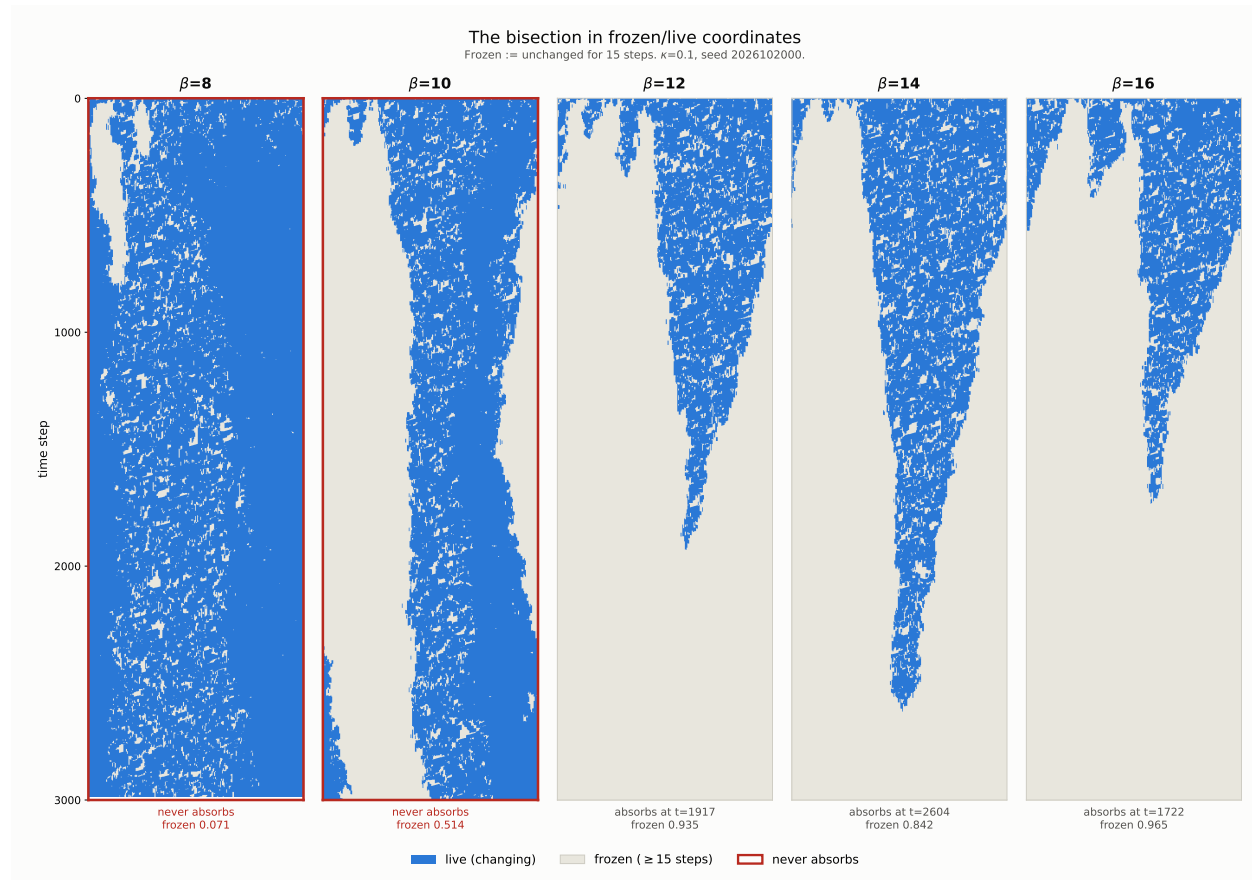


Figure 19

The bisection in frozen/live coordinates, $\kappa = 0.1$. Frozen is defined as unchanged for fifteen steps; at a four-step threshold the median frozen episode is itself four steps and the turnover count inflates thirteen-fold. $\beta = 12, 14$ and 16 taper to an absorbing state; $\beta = 8$ and 10 do not.

References

- Anthropic. (2026a). *Claude Opus*. Retrieved July 25, 2026, from <https://www.anthropic.com/claude/opus>
- Anthropic. (2026b). *Introducing Claude Fable 5*. Retrieved July 27, 2026, from <https://platform.claude.com/docs/en/about-claude/models/introducing-claude-fable-5>
- Aracena, J. (2008). Maximum number of fixed points in regulatory Boolean networks. *Bulletin of Mathematical Biology*, 70(5), 1398–1409.
- Corneli, J., & Maclean, E. (2015). The search for computational intelligence. In Y. J. Erden, R. Giovagnoli, & G. Dodig-Crnkovic (Eds.), *Social Aspects of Cognition and Computing Symposium, proceedings of the annual convention of the Society for the Study of Artificial Intelligence and Simulation of Behaviour (SSAISB)*. <https://arxiv.org/abs/1502.00130>
- Denmunzio, A., Formenti, E., & Provillard, J. (2011). Non-uniform cellular automata: Classes, dynamics, and decidability [arXiv:1107.5228].
- Langton, C. G. (1990). Computation at the edge of chaos: Phase transitions and emergent computation. *Physica D: Nonlinear Phenomena*, 42(1–3), 12–37. [https://doi.org/10.1016/0167-2789\(90\)90064-V](https://doi.org/10.1016/0167-2789(90)90064-V)
- Mitchell, M., Hraber, P. T., & Crutchfield, J. P. (1993). Revisiting the edge of chaos: Evolving cellular automata to perform computations. *Complex Systems*, 7(2), 89–130.
- Mori, T., Kudo, K., Namagawa, Y., Nakamura, R., Yamakawa, O., Suzuki, H., & Uesugi, T. (1998). Edge of chaos in rule-changing cellular automata. *Physica D: Nonlinear Phenomena*, 116, 275–282.
- OEIS Foundation Inc. (2011). Sequence A001818: Squares of double factorials; number of permutations of S_{2n} in which all cycles have even length [<https://oeis.org/A001818>; even-cycle interpretation dated 2011].
- OpenAI. (2026). *Codex*. Retrieved July 25, 2026, from <https://openai.com/codex/>

- Packard, N. H. (1988). Adaptation toward the edge of chaos. In J. A. S. Kelso, A. J. Mandell, & M. F. Shlesinger (Eds.), *Dynamic patterns in complex systems* (pp. 293–301). World Scientific.
- Pavlic, T. P., Adams, A. M., Davies, P. C. W., & Walker, S. I. (2014). Self-referencing cellular automata: A model of the evolution of information control in biological systems [arXiv:1405.4070].
- Riordan, J. (1968). *Combinatorial identities*. Wiley.
- Sakai, S., & Kanno, M. (2002). Structure of rule table and phase diagram of one dimensional cellular automata [arXiv:nlin/0204045].
- Schaller, M., & Svozil, K. (2025). Irreducible rules and equivalence classes of one-dimensional cellular automata [arXiv:2512.08117].
- Stanley, R. P. (1999). *Enumerative combinatorics, volume 2* [Problem 5.34(c)]. Cambridge University Press.
- Wuensche, A. (1999). Classifying cellular automata automatically: Finding gliders, filtering, and relating space-time patterns, attractor basins, and the Z parameter. *Complexity*, 4(3), 47–66.